Chapter 2
Nonparametric Statistics

"Nonparametric statistics" refers to statistics that do not require assumptions about parameters or do not depend on the distribution of the population. This is sometimes referred to as a "distribution-free test." The guidelines for selecting and using such methods are as follows:

1. When the sample size obtained from random sampling is small.
2. When the available data are measured on an ordinal scale or are in the form of ranked positions.
3. When the population distribution is unknown.
4. When the median represents the population more appropriately than the mean.
5. When there is uncertainty regarding the equality of variances in the population data.
6. When one wishes to test the independence between any two variables without involving parameter values.

Advantages

1. The calculation procedures are convenient and simple.
2. They can be applied to qualitative data.

Disadvantages

1. In some methods, the actual data used in the analysis are overlooked because they are converted into ranks or symbolic representations.
2. The accuracy of the tests is lower than that of parametric statistics.

Terms of Reference

2.1. Chi-square Tests

- Testing whether proportions follow the theoretically expected values (Tests of Proportions)
- Testing whether a distribution conforms to the expected form, also called the Goodness-of-Fit Test

- Testing the independence of two variables (Tests of Independence)

2.2. Hypothesis Testing concerning the Median of a Single Population
- Wilcoxon Signed Ranks Test

2.3. Hypothesis Testing concerning Two Related Populations
- Wilcoxon Signed Ranks Test with Paired Data

2.4. Hypothesis Testing concerning Two Independent Populations
- Mann–Whitney U–Test

2.5. Hypothesis Testing concerning Three or More Independent Populations
- Kruskal–Wallis H-Test

2.6. Rank Correlation Coefficient using Spearman's Method

Exercises for Chapter 2 and Solutions

Note: In this context, only two-tailed hypothesis tests will be considered.

2.1. Chi-Square Test

This refers to a hypothesis test used with data that are classified into categories, by considering the number (or frequency) of observed values (observed frequency) in each of those categories. Sometimes data that are expressed as numbers or frequencies are referred to as "count data."

2.1.1. The data used in the chi-square test can be classified into two types as follows:

2.1.1.1. **<u>One-way classified data</u>**, or data in the form of a "one-factor table" such as:

- A table showing the number of students who received an F grade in the course Statistics in Everyday Life, classified by faculty

Semester 1/2016

Faculty	Amount
Architecture and Design	15
Industrial Education	5
Industrial Technology and Management	10
Total	30

<u>The data contain a single factor</u>, namely the faculty, which is divided into 3 levels (or 3 categories). In this case, k = 3, where k represents <u>the number of levels or the number of categories</u>.

- The tests used with one-way classified data are as follows:
  1) Testing whether the proportions of the classified groups conform to the theoretically expected values, called "Tests of Proportions"
  2) Testing whether a distribution fits the expected form, called the "Goodness-of-Fit Test", such as:
     - Used to test the binomial distribution
     - Used to test the Poisson distribution
     - Used to test the normal distribution

The χ^2-Test (Formula 1) will be used to test whether the proportions in each category conform to the expected values.

2.1.1.2. Two-way classified data, or data in the form of a "contingency table," such as:

- A table showing the number of students who received an F grade in the course Statistics in Everyday Life

classified by faculty and gender for Semester 1/2016

Faculty	Gender		Total
	Male	Female	
Architecture and Design	10	5	15
Industrial Education	3	2	5
Industrial Technology and Management	2	8	10
Total	15	15	30

<u>The data contain two factors, namely faculty and gender.</u>
The first factor, faculty, has 3 levels, r = 3 (where r represents the number of rows)
The second factor, gender, has 2 levels, c = 2 (where c represents the number of columns)

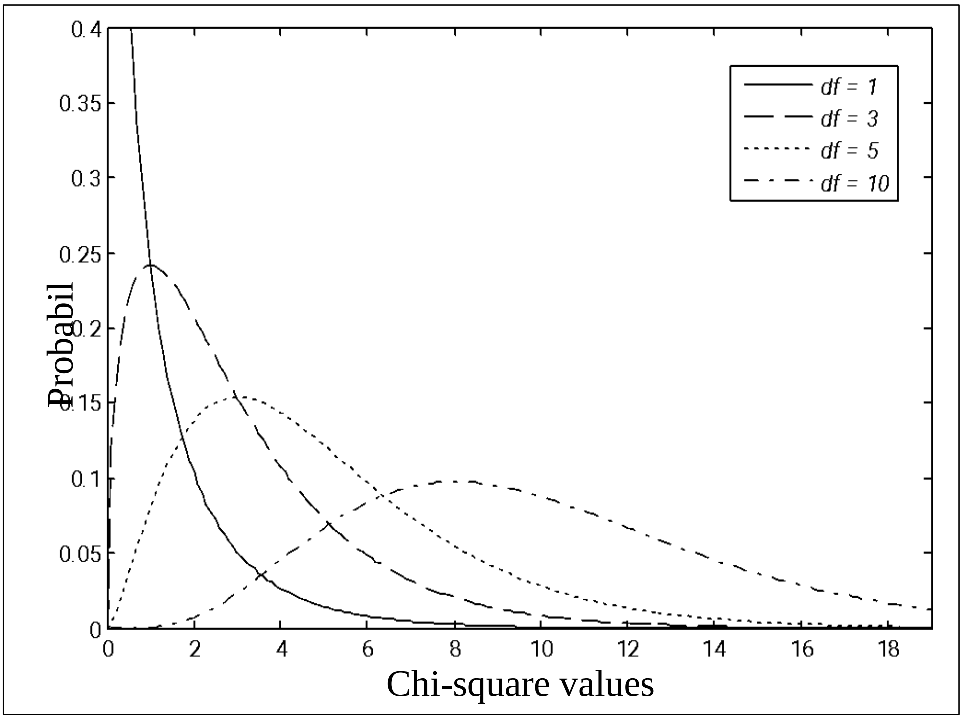
- The tests used with two-way classified data are as follows:
  1) Testing the independence of any two characteristics in the population, called "Tests of Independence."
  2) Testing homogeneity, called "Tests of Homogeneity."

The χ^2-Test (Formula 2) will be used to test the independence of the two factors.

2.1.2. **<u>Properties of the Chi-square Distribution</u>** (χ^2)

1) A chi-square random variable is a continuous random variable and takes values in the interval $[0,\infty]$.

2) A chi-square random variable has a mean equal to the degrees of freedom and a variance equal to twice the degrees of freedom, where degrees of freedom (d.f.) refers to the degree of independence.
3) <u>The distribution curve of a chi-square random variable is right-skewed</u> and becomes approximately symmetric as the number of degrees of freedom increases.



A figure illustrating the chi-square distribution at various levels of degrees of freedom.

2.1.3. <u>Hypothesis Testing for One-way Classified Data</u>

2.1.3.1. <u>Testing whether proportions (probabilities) conform to the specified theoretical values</u>

Let the characteristic of interest consist of a total of k categories.

Characteristic of interest category i	frequency (amount) O_i	theoretical probability π_i	expected frequency $E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
category 1	O_1	π_1	$E_1 = n\pi_1$	$\frac{(O_1 - E_1)^2}{E_1}$
category 2	O_2	π_2	$E_2 = n\pi_2$	$\frac{(O_2 - E_2)^2}{E_2}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
category k	O_k	π_k	$E_k = n\pi_k$	$\frac{(O_k - E_k)^2}{E_k}$
Total	$\sum_{i=1}^{k} O_i = n$	$\sum_{j=1}^{k} \pi_j = 1$	$\sum_{i=1}^{k} E_i = n = \sum_{i=1}^{k} O_i$	$\sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$

1) Formulating the hypotheses for the test

$H_0 : \pi_1 = \pi_{01}, \pi_2 = \pi_{02}, \ldots, \pi_k = \pi_{0k}$

$H_1 :$ There is at least one value that differs from the theoretical value specified in the hypothesis. H_0

Let π_i be the theoretical probability of the occurrence of the characteristic of interest in category i ; $i = 1, 2, \ldots, k$

2) Specify the significance level of the test, equal to α

3) The test statistic is:

$$\chi^2_{cal} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$
$$= \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \ldots + \frac{(O_k - E_k)^2}{E_k} \quad \text{(Formula 1)}$$

Where O_i is the observed frequency in category i ; $i = 1, 2, \ldots, k$

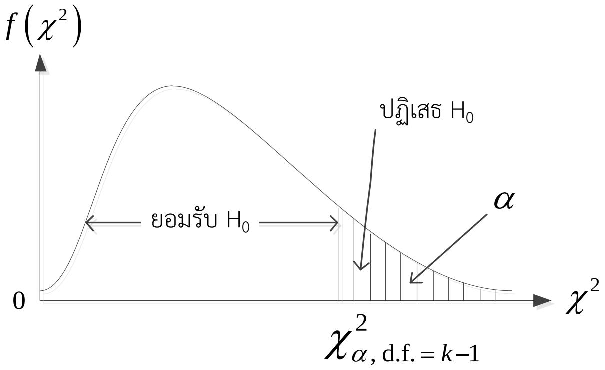
E_i is the expected frequency in category i ; $E_i = n\pi_i$

k is the number of categories or the number of classes.

n is the total frequency or the total sample size obtained.

$$\sum_{i=1}^{k} \mathrm{E}_i = n = \sum_{i=1}^{k} O_i$$

4) Decision Making



- <u>Reject</u> H_0 and <u>accept</u> H_1 If $\chi^2_{cal} > \chi^2_{\alpha,\mathrm{d.f.=k-1}}$
- <u>Accept</u> H_0 If $\chi^2_{cal} \le \chi^2_{\alpha,\mathrm{d.f.=k-1}}$

<u>Note</u>: If any class has $\mathrm{E}_i = n\pi_i < 5$ the data should be adjusted by combining two adjacent classes (by combining the O_i and E_i values of the two adjacent classes).

- **<u>Example of testing whether proportions (percentages) conform to the expected theoretical values</u>**

<u>Example 2.1</u> The Department of Statistics at a certain university has established the grading criteria for each course in the department as follows: Grade A 10%, Grade B 20%, Grade C 40%, Grade D 20%, and Grade F 10%. In Semester 1/2016, the instructor of the course Statistics in Everyday Life assigned letter grades to 100 students enrolled in this course as follows:

Grade	Amount O_i	π_i	$E_i = n\pi_i$	$\frac{(O_i-E_i)^2}{E_i}$
A	$O_1 = 12$	$\pi_1 = \frac{10}{100}$	$E_1 = n\pi_1 = 100\times\frac{10}{100} = 10$	$\frac{(O_1-E_1)^2}{E_1} = \frac{(12-10)^2}{10} = 0.4$
B	$O_2 = 18$	$\pi_2 = \frac{20}{100}$	$E_2 = n\pi_2 = 100\times\frac{20}{100} = 20$	$\frac{(O_2-E_2)^2}{E_2} = \frac{(18-20)^2}{20} = 0.2$
C	$O_3 = 45$	$\pi_3 = \frac{40}{100}$	$E_3 = n\pi_3 = 100\times\frac{40}{100} = 40$	$\frac{(O_3-E_3)^2}{E_3} = \frac{(45-40)^2}{40} = 0.625$
D	$O_4 = 15$	$\pi_4 = \frac{20}{100}$	$E_4 = n\pi_4 = 100\times\frac{20}{100} = 20$	$\frac{(O_4-E_4)^2}{E_4} = \frac{(15-20)^2}{20} = 1.25$
F	$O_5 = 10$	$\pi_5 = \frac{10}{100}$	$E_5 = n\pi_5 = 100\times\frac{10}{100} = 10$	$\frac{(O_5-E_5)^2}{E_5} = \frac{(10-10)^2}{10} = 0$
Total	$\sum_{i=1}^{5} O_i = 100$	$\sum_{i=1}^{5} \pi_i = 1$	$\sum_{i=1}^{5} E_i = 100 = \sum_{i=1}^{5} O_i$	$\chi^2_{cal} = \sum_{i=1}^{5}\frac{(O_i-E_i)^2}{E_i} = 2.475$

We would like to know whether this instructor's grading is consistent with the criteria specified by the department at $\alpha = 0.01$

Solution

1) Formulate the hypotheses for the test

H_0: $\pi_1 = \frac{10}{100}, \pi_2 = \frac{20}{100}, \pi_3 = \frac{40}{100}, \pi_4 = \frac{20}{100}, \pi_5 = \frac{10}{100}$

(This instructor's grading is consistent with the criteria specified by the department.)

H_1: There is at least one π_i value that differs from the value specified in

H_0; $i = 1, 2, 3, 4, 5$

(This instructor's grading is not consistent with the criteria specified by the department.)

2) Specify the significance level of the test, equal to $\alpha = 0.01$

3) The test statistic is:
$$\chi^2_{cal} = \sum_{i=1}^{5} \frac{(\mathrm{O}_i - \mathrm{E}_i)^2}{\mathrm{E}_i}$$
$$= 0.4 + 0.2 + 0.625 + 1.25 + 0$$
$$= 2.475$$

4) The critical region (or the value from the statistical table) is: Reject H_0 if $\chi^2_{cal} > \chi^2_{\alpha=0.01,\mathrm{d.f.}=5-1=4} = 13.2767$

5) Decision (accept H_0 or reject H_0): Accept H_0

6) Conclusion: This instructor's grading is consistent with the criteria specified by the department at the 0.01 significance level.

Example 2.2 From the population census conducted in the past 10 years in Nonthaburi Province, the following results were summarized:

20% of the population residing in Nonthaburi Province are under 16 years old.
20% of the population residing in Nonthaburi Province are between 16–30 years old.
25% of the population residing in Nonthaburi Province are between 31–45 years old.
15% of the population residing in Nonthaburi Province are between 46–60 years old.
20% of the population residing in Nonthaburi Province are over 60 years old.

The researcher would like to know whether the current age structure of the residents in Nonthaburi Province has changed from what was previously surveyed. Therefore, a sample of 200 residents in Nonthaburi Province was selected, and their ages were collected as shown in the table. Please assist the researcher in conducting the test at

$\alpha = 0.05$

Solution

Characteristic of interest i (age)	Frequency (amount) (O_i)	theoretical probability (π_i)	expected frequency $E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
under 16 years	$O_1 = 30$	$\pi_1 = \frac{20}{100}$	$E_1 = n\pi_1 = 200 \times \frac{20}{100} = 40$	$\frac{(O_1 - E_1)^2}{E_1} = \frac{(30-40)^2}{40} = 2.50$
16–30 years	$O_2 = 31$	$\pi_2 = \frac{20}{100}$	$E_2 = n\pi_2 = 200 \times \frac{20}{100} = 40$	$\frac{(O_2 - E_2)^2}{E_2} = \frac{(31-40)^2}{40} = 2.03$
31–45 years	$O_3 = 37$	$\pi_3 = \frac{25}{100}$	$E_3 = n\pi_3 = 200 \times \frac{25}{100} = 50$	$\frac{(O_3 - E_3)^2}{E_3} = \frac{(37-50)^2}{50} = 3.38$
46–60 years	$O_4 = 50$	$\pi_4 = \frac{15}{100}$	$E_4 = n\pi_4 = 200 \times \frac{15}{100} = 30$	$\frac{(O_4 - E_4)^2}{E_4} = \frac{(50-30)^2}{30} = 13.33$
over 60 years	$O_5 = 52$	$\pi_5 = \frac{20}{100}$	$E_5 = n\pi_5 = 200 \times \frac{20}{100} = 40$	$\frac{(O_5 - E_5)^2}{E_5} = \frac{(52-40)^2}{40} = 3.60$
Total	$\sum_{i=1}^{5} O_i = n = 200$	$\sum_{i=1}^{5} \pi_i = 1$	$\sum_{i=1}^{5} E_i = 200 = \sum_{i=1}^{5} O_i$	$\chi^2_{cal} = \sum_{i=1}^{5} \frac{(O_i - E_i)^2}{E_i} = 24.84$

1) Formulate the hypotheses for the test

$H_0:\ \pi_1 = 0.20,\ \pi_2 = 0.20,\ \pi_3 = 0.25,\ \pi_4 = 0.15,\ \pi_5 = 0.20$

(The current age structure of the residents in Nonthaburi Province has not changed from the previous survey.)

$H_1:$ There is at least one π_i value that differs from the value specified in H_0; $i = 1, 2, 3, 4, 5$

(The current age structure of the residents in Nonthaburi Province has changed from the previous survey.)

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) The test statistic is:

$$\chi^2_{cal} = \sum_{i=1}^{5} \frac{(O_i - E_i)^2}{E_i}$$
$$= 2.50 + 2.03 + 3.38 + 13.33 + 3.60$$
$$= 24.84$$

4) The critical region (or the value from the statistical table) is:
   Reject H_0 if $\chi^2_{cal} > \chi^2_{\alpha=0.05,\text{d.f.}=5-1=4} = 9.48773$
5) Decision (accept H_0 or reject H_0): reject H_0 and accept H_1
6) Conclusion: The current age structure of the residents in Nonthaburi Province has changed from the previous survey at the 0.05 significance level.
7)

Example 2.3 In an experiment of tossing a single die 600 times, it was found that the die showed faces 1, 2, 3, 4, 5, and 6 a total of 70, 80, 64, 120, 136, and 130 times, respectively. We would like to know whether this die is fair at $\alpha = 0.05$ (The value of π is not given directly but must be obtained from the theoretical result of the experiment.)

Solution

Characteristic of interest, category i	frequency (Amount) O_i	theoretical probability π_i	expected frequency $E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
1	$O_1 = 70$	$\pi_1 = \frac{1}{6}$	$E_1 = n\pi_1 = 600 \times \frac{1}{6} = 100$	$\frac{(O_1 - E_1)^2}{E_1} = \frac{(70-100)^2}{100} = 9$
2	$O_2 = 80$	$\pi_2 = \frac{1}{6}$	$E_2 = n\pi_2 = 600 \times \frac{1}{6} = 100$	$\frac{(O_2 - E_2)^2}{E_2} = \frac{(80-100)^2}{100} = 4$
3	$O_3 = 64$	$\pi_3 = \frac{1}{6}$	$E_3 = n\pi_3 = 600 \times \frac{1}{6} = 100$	$\frac{(O_3 - E_3)^2}{E_3} = \frac{(64-100)^2}{100} = 12.96$
4	$O_4 = 120$	$\pi_4 = \frac{1}{6}$	$E_4 = n\pi_4 = 600 \times \frac{1}{6} = 100$	$\frac{(O_4 - E_4)^2}{E_4} = \frac{(120-100)^2}{100} = 4$
5	$O_5 = 136$	$\pi_5 = \frac{1}{6}$	$E_5 = n\pi_5 = 600 \times \frac{1}{6} = 100$	$\frac{(O_5 - E_5)^2}{E_5} = \frac{(136-100)^2}{100} = 12.96$
6	$O_6 = 130$	$\pi_6 = \frac{1}{6}$	$E_6 = n\pi_6 = 600 \times \frac{1}{6} = 100$	$\frac{(O_6 - E_6)^2}{E_6} = \frac{(130-100)^2}{100} = 9$
Total	$\sum_{i=1}^{6} O_i = n = 600$	$\sum_{i=1}^{6} \pi_i = 1$	$\sum_{i=1}^{6} E_i = 600 = \sum_{i=1}^{6} O_i$	$\chi^2_{cal} = \sum_{i=1}^{6} \frac{(O_i - E_i)^2}{E_i} = 51.92$

1) Formulate the hypotheses for the test

H_0: $\pi_1 = 1/6, \pi_2 = 1/6, \pi_3 = 1/6, \pi_4 = 1/6, \pi_5 = 1/6, \pi_6 = 1/6$
(Fair dice)

H_1: There is at least one π_i value that differs from the value specified in

H_0; $i = 1, 2, 3, 4, 5, 6$
(Unfair dice)

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) 3) The test statistic is: $\chi^2_{cal} = \sum_{i=1}^{6} \frac{(O_i - E_i)^2}{E_i}$

$$= 9 + 4 + 12.96 + 4 + 12.96 + 9$$

$$= 51.92$$

4) The critical region (or the value from the statistical table) is:
Reject H_0 if $\chi^2_{cal} > \chi^2_{0.05, k-1=6-1=5} = 11.0705$

5) Decision (accept H_0 or reject H_0): reject H_0 and accept H_1

6) Conclusion: This die is not fair at the 0.05 significance level.

Example 2.4 According to a certain biological theory, it is found that breeding peas using method M yields the following ratio among the types: peas with round, yellow seeds : peas with round, green seeds : peas with wrinkled, yellow seeds : peas with wrinkled, green seeds = 5 : 2 : 2 : 1. If the result of breeding peas by this method for 100 plants produces the respective types in the following numbers: 51, 22, 19, and 8 plants, can we conclude that this theory is valid at the 0.01 significance level

Solution

Characteristic of interest i (Bean characteristics)	Frequency (amount) (O_i)	theoretical probability (π_i)	expected frequency $E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
(1) round + yellow	$O_1 = 51$	$\pi_1 = \frac{5}{10}$	$E_1 = n\pi_1 = 100 \times \frac{5}{10} = 50$	$\frac{(O_1 - E_1)^2}{E_1} = \frac{(51-50)^2}{50} = 0.02$
(2) round + green	$O_2 = 22$	$\pi_2 = \frac{2}{10}$	$E_2 = n\pi_2 = 100 \times \frac{2}{10} = 20$	$\frac{(O_2 - E_2)^2}{E_2} = \frac{(22-20)^2}{50} = 0.20$
(3) wrinkled + yellow	$O_3 = 19$	$\pi_3 = \frac{2}{10}$	$E_3 = n\pi_3 = 100 \times \frac{2}{10} = 20$	$\frac{(O_3 - E_3)^2}{E_3} = \frac{(19-20)^2}{50} = 0.05$
(4) wrinkled + green	$O_4 = 8$	$\pi_4 = \frac{1}{10}$	$E_4 = n\pi_4 = 100 \times \frac{1}{10} = 10$	$\frac{(O_4 - E_4)^2}{E_4} = \frac{(8-10)^2}{10} = 0.40$
Total	$\sum_{i=1}^{4} O_i = n = 100$	$\sum_{i=1}^{4} \pi_i = 1$	$\sum_{i=1}^{4} E_i = 100 = \sum_{i=1}^{4} O_i$	$\chi^2_{cal} = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i} = 0.67$

1) Formulate the hypotheses for the test

$H_0:\ \pi_1 = \frac{5}{10}, \pi_2 = \frac{2}{10}, \pi_3 = \frac{2}{10}, \pi_4 = \frac{1}{10}$ (The theory is valid.)

$H_1:$ There is at least one π_i value that differs from the value specified in

H_0; $i = 1, 2, 3, 4$ (The theory is not valid.)

2) Specify the significance level of the test, equal to $\alpha = 0.01$

3) The test statistic is: $\chi^2_{cal} = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$

$$= \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \frac{(O_3 - E_3)^2}{E_3} + \frac{(O_4 - E_4)^2}{E_4}$$

$$= 0.02 + 0.20 + 0.05 + 0.40$$

$$= 0.67$$

4) The critical region (or the value from the statistical table) is:
   Reject H_0
   if $\chi^2_{cal} > \chi^2_{0.01,k-1=4-1=3} = 11.3449$
5) Decision (accept H_0 or reject H_0) : accept H_0
6) Conclusion: The theory of pea breeding by method M is valid at the 0.01 significance level.

<u>Example 2.5</u> From a survey of preferences for candidates running for Member of Parliament (MP) in Bangkok, Constituency 4, there are 5 candidates, namely Phumin, Charoemchai, Atthawit, Lina, and Prawit. The researcher randomly selected a sample of 1,800 eligible voters in that constituency and asked which candidate they liked the most. The data obtained are as follows:

Candidates	Number of Respondents by Preferred Candidate (O_i)	π_i	$E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
Phumin	$O_1 = 360$	$\pi_1 = \frac{1}{6}$	$E_1 = n\pi_1 = 1800 \times \frac{1}{6} = 300$	$\frac{(O_1 - E_1)^2}{E_1} = \frac{(360-300)^2}{300} = 12$
Chaloemchai	$O_2 = 300$	$\pi_2 = \frac{1}{6}$	$E_2 = n\pi_2 = 1800 \times \frac{1}{6} = 300$	$\frac{(O_2 - E_2)^2}{E_2} = \frac{(300-300)^2}{300} = 0$
Atthawit	$O_3 = 630$	$\pi_3 = \frac{2}{6} = \frac{1}{3}$	$E_3 = n\pi_3 = 1800 \times \frac{1}{3} = 600$	$\frac{(O_3 - E_3)^2}{E_3} = \frac{(630-600)^2}{600} = 1.5$
Lina	$O_4 = 270$	$\pi_4 = \frac{1}{6}$	$E_4 = n\pi_4 = 1800 \times \frac{1}{6} = 300$	$\frac{(O_4 - E_4)^2}{E_4} = \frac{(270-300)^2}{300} = 3$
Prawit	$O_5 = 240$	$\pi_5 = \frac{1}{6}$	$E_5 = n\pi_5 = 1800 \times \frac{1}{6} = 300$	$\frac{(O_5 - E_5)^2}{E_5} = \frac{(240-300)^2}{300} = 12$
Total	$\sum_{i=1}^{5} O_i = 1800$	$\sum_{i=1}^{5} \pi_i = 1$	$\sum_{i=1}^{5} E_i = 1800 = \sum_{i=1}^{5} O_i$	$\chi^2_{cal} = \sum_{i=1}^{5} \frac{(O_i - E_i)^2}{E_i} = 28.5$

The researcher believes that Mr. Atthawit is twice as popular as the other candidates. Please test this belief and state the conclusion at the 0.01 significance level.

Solution

1) Formulate the hypotheses for the test

$H_0:\ \pi_1 = 1/6, \pi_2 = 1/6, \pi_3 = 2/6, \pi_4 = 1/6, \pi_5 = 1/6$

(Mr. Atthawit is twice as popular as the other candidates.)

$H_1:$ There is at least one π_i value that differs from the value specified in

$H_0;\quad i = 1, 2, 3, 4, 5$

(Mr. Atthawit is not twice as popular as the other candidates.)

2) Specify the significance level of the test, equal to $\alpha = 0.01$

3) The test statistic is: $\chi^2_{cal} = \sum_{i=1}^{5} \frac{(O_i - E_i)^2}{E_i}$

$$= 12 + 0 + 1.5 + 3 + 16.5$$

$$= 33$$

4) The critical region (or the value from the statistical table) is:

   Reject H_0

   If $\chi^2_{cal} > \chi^2_{\alpha=0.01,\text{d.f.}=5-1=4} = 13.2767$

5) Decision (accept H_0 or reject H_0): reject H_0 and accept H_1
6) Conclusion: Mr. Atthawit is not twice as popular as the other candidates at the 0.01 significance level.

2.1.3.2. <u>Goodness-of-Fit Test</u>

The Goodness-of-Fit Test is used to assess whether a distribution conforms to the expected distribution. The expected distribution may be a continuous distribution, such as the normal distribution, or a discrete distribution, such as the binomial distribution or the Poisson distribution. The hypotheses are as follows:

H_0: The population has a distribution of type Y

H_1: The population does not have a distribution of type Y

where Y may be normal, binomial, or Poisson.

The test statistic is
$$\chi^2_{cal} = \sum_{i=1}^{k} \frac{(\mathrm{O}_i - \mathrm{E}_i)^2}{\mathrm{E}_i}$$

When χ^2 = is the symbol for the chi-square statistic

O_i = is the observed frequency

E_i = is the expected (or specified) frequency

n = is the total frequency or the total sample size obtained

π_i = is the probability of occurrence in category i under the expected distribution

Formula is $\mathrm{E}_i = n \times \pi_i$

The statistic χ^2 follows a chi-square distribution with degrees of freedom (k-1)-m m is the number of parameters that must be estimated. Its value depends on the expected distribution being tested. For example, if the hypothesis is $H_0:$ the population follows a normal distribution

m= 2 because the mean and the variance must be estimated $\chi^2 \sim \chi^2_{(k-1-2)}$

If the hypothesis is $H_0:$ the population follows a normal distribution with mean = 5 and variance = 6

m = 0 because neither the mean (m) and the variance (s^2) $\chi^2 \sim \chi^2_{(k-1)}$

** Note: If any class has an expected frequency of less than 5, it should be combined with an adjacent class.

Example 2.6 From 100 water samples collected from a certain river for analysis, to determine the number of living organisms found in each water sample, the data can be summarized as follows.

Number of living organisms	0	1	2	3	4	5	6	7	Total
Frequency	15	30	25	20	5	4	1	0	100

At the 0.05 significance level, test whether the distribution of the number of living organisms follows a Poisson distribution.

Solution

1) Formulate the hypotheses for the test

The hypotheses are

$H_0:$ The distribution of the number of living organisms follows a Poisson distribution.

$H_1:$ the distribution of the number of living organisms does not follow a Poisson distribution.

2) The probability mass function of the Poisson distribution is

$$\pi = f(x) = \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} \; ; \quad x = 0,1,2,\ldots$$

In this case, the parameter λ is unknown, so it is estimated by the sample mean.

$$\bar{x} = \frac{15(0)+30(1)+25(2)+20(3)+4(5)+5(4)+6(1)+7(0)}{100} = 1.85$$

Number of living organisms (x_i)	O_i	$\pi_i = \dfrac{e^{-\lambda}\lambda^{x_i}}{x_i!}$	$E_i = n\pi_i$
0	15	0.156	15.567
1	30	0.290	28.955
2	25	0.269	26.928
3	20	0.167	16.696
4	5	0.078	7.763
5	4	0.029	2.888
6	1	0.009	0.895
7	0	0.002	0.238
Total	100	1.0000	

*** Since there are two classes with frequencies less than 5, they are combined with adjacent classes.

Number of living organisms (x_i)	O_i	$E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
0	15	15.567	0.021
1	30	28.955	0.038
2	25	26.928	0.138
3	20	16.696	0.654
4	10	11.785	0.271
Total	100		1.121

Specify the significance level of the test $\alpha = 0.05$

The test statistic is: $\chi^2_{cal} = \sum_{i=1}^{5} \frac{(O_i - E_i)^2}{E_i}$

$$= \frac{(15-15.567)^2}{15.567} + \frac{(30-28.955)^2}{28.955} + \ldots = 4.756$$

5) The critical region (or statistics table values) From the χ^2 table, we obtain $\chi^2_{\alpha,\text{d.f.}}$ = $\chi^2_{0.05,(5-1)-1=3}$ = 7.815 and will reject the hypothesis H_0 when $\chi^2_{cal} > \chi^2_{\alpha,\text{d.f.}}$

6) Decision (accept H_0 or reject H_0): accept H_0

7) Conclusion: Since the value of χ^2_{cal} = 1.121 < 7.815 we therefore accept the main hypothesis H_0 that is, the distribution of the number of

living organisms follows a Poisson distribution at the 0.05 significance level.

Example 2.7 From interviewing 83 Thai people in the Bang Sue district about their monthly income, the following data were obtained:

Monthly income(unit:100,000baht)	Number(peopl
Less than 0.10	12
0.10 to less than 0.15	20
0.15 to less than 0.20	23
0.20 to less than 0.25	15
0.25 and above	13
Total	83

Test whether the income of people in the Bang Sue district follows a normal distribution with a mean monthly income of 20,000 baht and a standard deviation of 5,000 baht at the 99% confidence level.

1) The hypotheses for the test are:

$H_0:$ The income of people in the Bang Sue district follows a normal distribution with mean monthly income of 20,000 baht and standard deviation of 5,000 baht.

$H_1:$ The income of people in the Bang Sue district does not follow a normal distribution with mean monthly income 20,000 baht and standard deviation 5,000 baht.

2) Specify the significance level of the test, equal to $\alpha = 0.01$
3) The test statistic is:

$$\chi^2_{cal} = \sum_{i=1}^{k} \frac{(\mathrm{O}_i - \mathrm{E}_i)^2}{\mathrm{E}_i}$$

When $O_1=12$, $O_2 = 20$, $O_3 = 23$, $O_4 = 15$, $O_5 = 13$

Let x be the monthly income of people in the Bang Sue district.

Compute the E_i as follows,

Assuming H_0 is true $\mathrm{E}_i = n\pi_i$

Where π_i = the probability that a person in the Bang Sue district has income in interval i ; i =1,2,3,4,5

That is, compute π_i when $x \sim N(\mu = 0.2, \sigma = 0.05)$

Convert x to z, where $z = \frac{x-\mu}{\sigma} = \frac{x_i - 0.2}{0.05}$ to obtain the value of z and then obtain the E_i as shown in the following table.

class	income (x)	$z = \frac{x_i - 0.2}{0.05}$	z	π_i	$E_i = 83 \times \pi_i$	O_i
1	$x < 0.1$	$\frac{0.1-0.2}{0.05} = -2$	$z < -2$	0.0228	1.89	12
2	$0.1 < x < 0.15$	$\frac{0.15-0.2}{0.05} = -1$	$-2 < z < -1$	0.1359	11.28	20
3	$0.15 < x < 0.2$	$\frac{0.2-0.2}{0.05} = 0$	$-1 < z < 0$	0.3413	28.33	23
4	$0.2 < x < 0.25$	$\frac{0.25-0.2}{0.05} = 1$	$0 < z < 1$	0.3413	28.33	15
5	$x \geq 0.25$		$z \geq 1$	0.1587	13.17	13

Since $E_1 = 1.89$, which is less than 5, the first- and second-income classes are therefore combined. Thus, there remain 4 classes, and $O_1 = 12+20 = 32$ $E_1 = 1.89+11.28 = 13.17$

$$\chi^2_{cal} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{(32-13.17)^2}{13.17} + \frac{(23-28.33)^2}{28.33} + \frac{(15-28.33)^2}{28.33} + \frac{(13-13.17)^2}{13.17}$$

$$= 34.20$$

4) The critical region (or statistics table values) is: Reject H_0 when $\chi^2_{cal} > \chi^2_{0.01,(4-1)}$ from the table χ^2 we obtain $\chi^2_{0.01,3} = 11.34$
5) Decision (accept H_0 or reject H_0): reject H_0
6) Conclusion: The monthly income of people in the Bang Sue district does not follow a normal distribution with a mean of

20,000 baht and a standard deviation of 5,000 baht at the 0.01 significance level.

For testing other types of distributions, the method of calculation is similar.

2.1.4. <u>Hypothesis Testing for Two-way Classified Data</u>

2.1.4.1. <u>Test of Independence of Two Variables</u>

This is a test to determine whether the data in the rows and columns of an $r \times c$ contingency table are related. The data in the horizontal direction are divided into r levels, and the data in the vertical direction are divided into c levels.

Classification B	Classification A				Total
	A_1	A_2	$\cdots$	A_c	
B_1	O_{11}	O_{12}	$\cdots$	O_{1c}	r_1
B_2	O_{21}	O_{22}	$\cdots$	O_{2c}	r_2
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
B_r	O_{r1}	O_{r2}	$\cdots$	O_{rc}	r_r
Total	c_1	c_2	$\cdots$	c_c	n

Where
$$\boxed{\sum_{i=1}^{r}\sum_{j=1}^{c} O_{ij} = \sum_{i=1}^{r} r_i = \sum_{i=1}^{c} c_i = n = \sum_{i=1}^{r}\sum_{j=1}^{c} \mathrm{E}_{ij}}$$

And E_{ij} **<u>is the expected number of people</u>** (or the expected frequency) **<u>in row i and column j</u>**

where r is the total number of rows

c is the total number of columns

O_{ij} is the observed frequency (or number) in row i and column j

r_i is the total frequency in row i ; $i = 1, 2, \ldots, r$

c_j is the total frequency in column j ; $j = 1, 2, \ldots, c$

n is the total number of observations

π_i is the theoretical probability of occurrence in category i ; $0 \le \pi_i \le 1$

<u>Steps in Hypothesis Testing for the Independence of Two Characteristics</u>

1) Formulating the hypotheses for the test

$H_0:$ The two characteristics under study are <u>independent</u>.

$H_1:$ The two characteristics under study are <u>not independent</u>.

2) Specify the significance level of the test, equal to α

3) The test statistic is

$$\chi^2_{cal} = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(\mathrm{O}_{ij}-\mathrm{E}_{ij})^2}{\mathrm{E}_{ij}}$$

$$= \frac{(\mathrm{O}_{11}-\mathrm{E}_{11})^2}{\mathrm{E}_{11}} + \frac{(\mathrm{O}_{12}-\mathrm{E}_{12})^2}{\mathrm{E}_{12}} + \ldots + \frac{(\mathrm{O}_{rc}-\mathrm{E}_{rc})^2}{\mathrm{E}_{rc}}$$

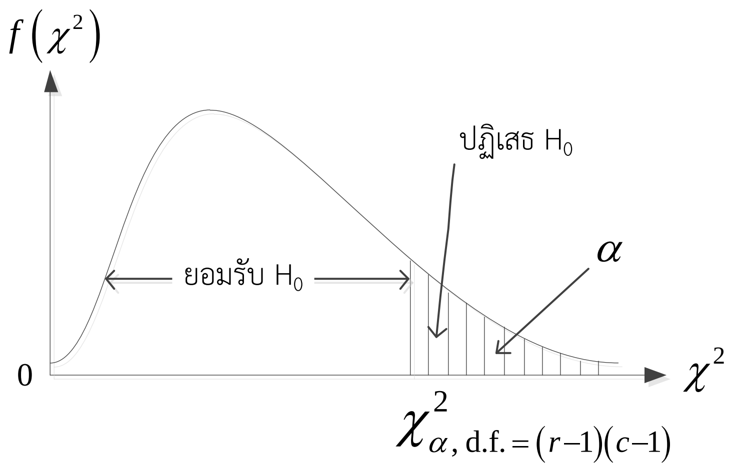
where E_{ij} is <u>the expected number of people in row i and column j</u>

$\mathrm{E}_{ij} = \dfrac{r_i \times c_j}{n}$ where r_i is the total number of people in row i

c_j is the total number of people in column j

such as $\mathrm{E}_{12} = \dfrac{r_1 \times c_2}{n}$ and $\mathrm{E}_{32} = \dfrac{r_3 \times c_2}{n}$ and so forth.

4) Decision



- <u>reject</u> H_0 and <u>accept</u> H_1 if $\chi^2_{cal} > \chi^2_{\alpha,\text{d.f.}=(r-1)(c-1)}$

- <u>accept</u> H_0 if $\chi^2_{cal} \le \chi^2_{\alpha,\text{d.f.}=(r-1)(c-1)}$

<u>Example 2.8</u> In a survey of the opinions of 120 residents of Nonthaburi Province regarding whether they agree or disagree with various projects under the new government's policies, the researcher collected data from the people in Nonthaburi Province and obtained the following results:

No.	gender	opinion	No.	gender	opinion	No.	gender	opinion	No.	gender	opinion
1	female	agree	31	female	disagree	61	female	agree	91	female	agree
2	female	agree	32	female	disagree	62	female	agree	92	female	agree
3	female	disagree	33	female	disagree	63	female	disagree	93	female	agree
4	female	disagree	34	female	disagree	64	female	disagree	94	female	agree
5	female	disagree	35	female	disagree	65	female	disagree	95	female	agree
6	female	disagree	36	female	disagree	66	female	disagree	96	female	agree
7	female	disagree	37	female	disagree	67	female	disagree	97	female	agree
8	female	disagree	38	female	disagree	68	female	disagree	98	female	disagree
9	female	disagree	39	female	agree	69	female	disagree	99	female	disagree
10	female	disagree	40	female	agree	70	female	disagree	100	female	disagree
11	female	disagree	41	female	agree	71	female	disagree	101	female	disagree
12	female	disagree	42	female	agree	72	female	disagree	102	female	disagree
13	female	disagree	43	female	disagree	73	female	disagree	103	male	disagree
14	female	disagree	44	female	disagree	74	female	disagree	104	male	disagree
15	female	disagree	45	female	disagree	75	female	disagree	105	male	disagree
16	female	disagree	46	female	disagree	76	female	disagree	106	male	disagree
17	female	disagree	47	female	disagree	77	female	disagree	107	male	disagree
18	female	disagree	48	female	disagree	78	female	disagree	108	male	disagree
19	female	disagree	49	female	disagree	79	male	disagree	109	male	disagree
20	female	disagree	50	female	disagree	80	male	disagree	110	male	disagree
21	female	disagree	51	female	disagree	81	male	disagree	111	male	disagree
22	female	disagree	52	female	disagree	82	female	disagree	112	female	disagree
23	female	disagree	53	female	disagree	83	female	disagree	113	female	disagree
24	female	disagree	54	female	disagree	84	female	disagree	114	female	disagree
25	female	disagree	55	female	disagree	85	female	disagree	115	female	disagree
26	male	disagree	56	female	disagree	86	female	disagree	116	female	disagree
27	male	disagree	57	female	disagree	87	female	disagree	117	female	disagree
28	male	disagree	58	female	disagree	88	female	disagree	118	female	disagree
29	male	disagree	59	female	agree	89	female	disagree	119	female	disagree
30	male	disagree	60	female	agree	90	female	disagree	120	female	disagree

1) From the data obtained by the researcher, construct a two-way classified table, also known as a contingency table.
2) Based on the given data, the researcher believes that "gender and the opinions of people in Nonthaburi Province are related." Do you agree or not? Show the method for testing the hypothesis and present the conclusion at the 0.01 significance level.

Solution

Contingency Table

Gender	Opinion		Total
	agree	disagree	
male	10	50	60
female	20	40	60
Total	30	90	120

1) Formulate the hypotheses for the test

H_0: Gender and the opinions of people in Nonthaburi Province are not related.

H_1: Gender and the opinions of people in Nonthaburi Province are related.

2) Specify the significance level of the test, equal to $\alpha = 0.01$
3) The test statistic used is

$$\chi^2_{cal} = \sum_{i=1}^{2}\sum_{j=1}^{2}\frac{(\mathrm{O}_{ij}-\mathrm{E}_{ij})^2}{\mathrm{E}_{ij}}$$

$$= \frac{(\mathrm{O}_{11}-\mathrm{E}_{11})^2}{\mathrm{E}_{11}} + \frac{(\mathrm{O}_{12}-\mathrm{E}_{12})^2}{\mathrm{E}_{12}} + \frac{(\mathrm{O}_{21}-\mathrm{E}_{21})^2}{\mathrm{E}_{21}} + \frac{(\mathrm{O}_{22}-\mathrm{E}_{22})^2}{\mathrm{E}_{22}}$$

$$= \frac{(10-15)^2}{15} + \frac{(50-45)^2}{45} + \frac{(20-15)^2}{15} + \frac{(40-45)^2}{45}$$

$$= 1.6667 + 0.5556 + 1.6667 + 0.5556$$

$$= 4.4446$$

4) The critical region (or statistics table values) is: Reject H_0 if $\chi^2_{cal} > \chi^2_{\alpha=0.01,\mathrm{d.f.}=(2-1)(2-1)=1} = 6.63$
5) Decision (accept H_0 or reject H_0): accept H_0
6) Conclusion: Gender and the opinions of people in Nonthaburi Province are not related at the 0.01 significance level.

Example 2.9 In a survey of the popularity of a certain political party among the population, the researcher randomly selected eligible voters from Chiang Mai, Songkhla, Khon Kaen, and Rayong Provinces in the numbers 190, 210, 195, and 205 persons, respectively, and obtained the following data. Test whether the popularity of this political party among the people depends on the province in which they reside at $\alpha = 0.05$

Solution

favoring	province				Total
	Chiang Mai	Songkhla	Khon Kaen	Rayong	
favored	$O_{11} = 70$	$O_{12} = 60$	$O_{13} = 55$	$O_{14} = 50$	$r_1 = 235$
unfavored	$O_{21} = 120$	$O_{22} = 150$	$O_{23} = 140$	$O_{24} = 155$	$r_2 = 565$
Total	$c_1 = 190$	$c_2 = 210$	$c_3 = 195$	$c_4 = 205$	$n = 800$

<u>Calculation of the expected number of people in row</u> i <u>and column</u> j $(\mathrm{E}_{ij}) = \dfrac{r_i \times c_j}{n}$

$$\mathrm{E}_{11} = \frac{r_1 \times c_1}{n} = \frac{235 \times 190}{800} = 55.8, \qquad \mathrm{E}_{12} = \frac{r_1 \times c_2}{n} = \frac{235 \times 210}{800} = 61.7,$$

$$\mathrm{E}_{13} = \frac{r_1 \times c_3}{n} = \frac{235 \times 195}{800} = 57.3, \qquad \mathrm{E}_{14} = \frac{r_1 \times c_4}{n} = \frac{235 \times 205}{800} = 60.2$$

$$\mathrm{E}_{21} = \frac{r_2 \times c_1}{n} = \frac{565 \times 190}{800} = 134.2, \qquad \mathrm{E}_{22} = \frac{r_2 \times c_2}{n} = \frac{565 \times 210}{800} = 148.3,$$

$$\mathrm{E}_{23} = \frac{r_2 \times c_3}{n} = \frac{565 \times 195}{800} = 137.7, \qquad \mathrm{E}_{24} = \frac{r_2 \times c_4}{n} = \frac{565 \times 205}{800} = 144.8$$

1) Formulate the hypotheses for the test

$H_0:$ The popularity of the political party among the people does not depend on the province in which they reside.

$H_1:$ The popularity of the political party among the people depends on the province in which they reside.

2) Specify the significance level of the test, equal to $\alpha = 0.05$.

3) The test statistic used is $\chi^2_{cal} = \displaystyle\sum_{i=1}^{2}\sum_{j=1}^{4}\frac{(\mathrm{O}_{ij} - \mathrm{E}_{ij})^2}{\mathrm{E}_{ij}}$

$$= \frac{(\mathrm{O}_{11} - \mathrm{E}_{11})^2}{\mathrm{E}_{11}} + \frac{(\mathrm{O}_{12} - \mathrm{E}_{12})^2}{\mathrm{E}_{12}} + \ldots + \frac{(\mathrm{O}_{24} - \mathrm{E}_{24})^2}{\mathrm{E}_{24}}$$

$$= \frac{(70-55.81)^2}{55.81} + \frac{(60-61.69)^2}{61.69} + \ldots + \frac{(155-144.78)^2}{144.78}$$

$$= 3.61 + 0.05 + 0.09 + 1.73 + 1.50 + 0.02 + 0.04 + 0.72$$

$$= 7.76$$

4) The critical region (or statistics table values) is: Reject H_0

If $\chi^2_{cal} > \chi^2_{0.05,\text{df}=(2-1)(4-1)=3} = 7.81473$

5) Decision (accept H_0 or reject H_0): Accept H_0

6) Conclusion: The popularity of the political party does not depend on the province in which the people reside at the 0.05 significance level.

Example 2.10 From a survey of 200 people, both smokers and non-smokers, the following data were obtained:

Among the 120 smokers, 16 people have lung cancer.

Among the 160 people who do not have lung cancer, 56 are non-smokers.

From this information, can we conclude that smoking influences lung cancer at the 0.05 significance level?

Solution

Lung cancer	Smoking behavior		Total
	Smokers	Non-smokers	
present	$O_{11} = 16$	$O_{12} = 24$	$r_1 = 40$
not present	$O_{21} = 104$	$O_{22} = 56$	$r_2 = 160$
Total	$c_1 = 120$	$c_2 = 80$	$n = 200$

$\text{E}_{11} = \frac{r_1 \times c_1}{n} = \frac{40 \times 120}{200} = 24,$ $\text{E}_{12} = \frac{r_1 \times c_2}{n} = \frac{40 \times 80}{200} = 16,$

$\text{E}_{21} = \frac{r_2 \times c_1}{n} = \frac{160 \times 120}{200} = 96,$ $\text{E}_{22} = \frac{r_2 \times c_2}{n} = \frac{160 \times 80}{200} = 64$

1) Formulate the hypotheses for the test

H_0 : Smoking has no effect on causing lung cancer.

H_1 : Smoking influences lung cancer.

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) The test statistic used is $\chi^2_{cal} = \sum_{i=1}^{2}\sum_{j=1}^{2}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}$

$$= \frac{(O_{11}-E_{11})^2}{E_{11}} + \frac{(O_{12}-E_{12})^2}{E_{12}} + \frac{(O_{21}-E_{21})^2}{E_{21}} + \frac{(O_{22}-E_{22})^2}{E_{22}}$$

$$= 2.67 + 4.00 + 0.67 + 1.00$$

$$= 8.34$$

4) The critical region (or statistics table values) is: Reject H_0 if $\chi^2_{cal} > \chi^2_{0.05,\mathrm{df}=(2-1)(2-1)=1} = 3.84146$

5) Decision (accept H_0 or reject H_0): Reject H_0 and accept H_1

6) Conclusion: Smoking influences lung cancer at the 0.05 significance level.

Example 2.11 A cosmetic company collected data on the gender and work experience of all 500 employees in the company, with the following results:

Gender	Work experience (years)		
	0 – 3	4 – 7	more than 7 years
Male	45	120	85
Female	55	130	65

If there are 150 employees with more than 7 years of work experience, test whether gender and work experience of employees in this company are independent at the 0.01 significance level.

Solution

1) Formulating the hypotheses for the test

H_0 : Gender and work experience of the employees are independent.

H_1 : Gender and work experience of the employees are not independent.

2) Specify the significance level of the test, equal to $\alpha = 0.01$
3) The test statistic used is $\chi^2_{cal} = \sum_{i=1}^{2}\sum_{j=1}^{3}\frac{(\mathrm{O}_{ij}-\mathrm{E}_{ij})^2}{\mathrm{E}_{ij}}$

$$= \frac{(45-50)^2}{50} + \frac{(120-125)^2}{125} + \frac{(85-75)^2}{75} + \ldots + \frac{(65-75)^2}{75}$$

$$= 0.5 + 0.2 + 1.33 + 0.5 + 0.2 + 1.33$$

$$= 4.06$$

$$\mathrm{E}_{11} = \frac{r_1 \times c_1}{n} = \frac{250\times 100}{500} = 50,\quad \mathrm{E}_{12} = \frac{r_1 \times c_2}{n} = \frac{250\times 250}{500} = 125,\quad \mathrm{E}_{13} = \frac{r_1 \times c_3}{n} = \frac{250\times 150}{500} = 75$$

$$\mathrm{E}_{21} = \frac{r_2 \times c_1}{n} = \frac{250\times 100}{500} = 50,\quad \mathrm{E}_{22} = \frac{r_2 \times c_2}{n} = \frac{250\times 250}{500} = 125,\quad \mathrm{E}_{23} = \frac{r_2 \times c_3}{n} = \frac{250\times 150}{500} = 75$$

4) The critical region (or statistics table values) is: Reject H_0 if $\chi^2_{cal} > \chi^2_{0.01,\mathrm{df}=(3-1)(2-1)=2} = 9.21034$
5) Decision (accept H_0 or reject H_0): Accept H_0
6) Conclusion: Gender and work experience of the employees are independent at the 0.05 significance level.

Example 2.12 In studying two characteristics of normal individuals, namely eye color and hair color, to determine whether they are related, the researcher randomly selected 100 normal individuals and recorded their eye color and hair color. The data obtained are as follows:

31 people have brown eyes and brown hair
21 people have brown eyes and black hair
14 people have black eyes and brown hair
34 people have black eyes and black hair

1. Test whether eye color and hair color are related at the 0.10 significance level.

Solution

Hair color	eye color		Total
	brown	black	
brown	$O_{11}=31$	$O_{12}=14$	$r_1=45$
black	$O_{21}=21$	$O_{22}=34$	$r_2=55$
Total	$c_1=52$	$c_2=48$	$n=100$

$$\mathrm{E}_{11}=\frac{r_1\times c_1}{n}=\frac{45\times 52}{100}=23.4, \qquad \mathrm{E}_{12}=\frac{r_1\times c_2}{n}=\frac{45\times 48}{100}=21.6$$

$$\mathrm{E}_{21}=\frac{r_2\times c_1}{n}=\frac{55\times 52}{100}=28.6, \qquad \mathrm{E}_{22}=\frac{r_2\times c_2}{n}=\frac{55\times 48}{100}=26.4$$

H_0: Eye color and hair color are not related.

H_1: Eye color and hair color are related.

2) Specify the significance level of the test, equal to $\alpha=0.10$

3) The test statistic used is $\chi^2_{cal}=\sum_{i=1}^{2}\sum_{j=1}^{2}\frac{(\mathrm{O}_{ij}-\mathrm{E}_{ij})^2}{\mathrm{E}_{ij}}$

$$=\frac{(31-23.4)^2}{23.4}+\frac{(21-28.6)^2}{28.6}+\frac{(14-21.6)^2}{21.6}+\frac{(34-36.4)^2}{36.4}$$

$$=2.47+2.67+2.02+2.19$$

$$=9.35$$

4) The critical region (or statistics table values) is: Reject H_0 if $\chi^2_{cal}>\chi^2_{0.10,\mathrm{df}=(2-1)(2-1)=1}=2.70554$

5) Decision (accept H_0 or reject H_0): Reject H_0 and accept H_1

6) Conclusion: Eye color and hair color are related at the 0.10 significance level.

2. If the researcher obtained the above sample data from City A, test whether the people in this city have eye color and hair color in the given proportions to approximately the same extent at the 0.05 significance level (this is a test of whether the proportions conform to the expected values).

Solution

Definition of π :

π_1 represents the probability that a person has brown eyes and brown hair $=1/4$

π_2 represents the probability that a person has brown eyes and black hair $=1/4$

π_3 represents the probability that a person has black eyes and brown hair $=1/4$

π_4 represents the probability that a person has black eyes and black hair $=1/4$

(since the proportions of people in all four characteristic groups are approximately equal, or in the ratio 1 : 1 : 1 : 1)

Characteristic of interest i (eye color and hair color)	Frequency (number) O_i	Theoretical probability π_i	Expected frequenc $E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
Brown eyes and brown hair	$O_1 = 31$	$\pi_1 = \frac{1}{4}$	$E_1 = n\pi_1 = 100 \times \frac{1}{4} = 25$	$\frac{(O_1 - E_1)^2}{E_1} = \frac{(31-25)^2}{25} = 1.44$
Brown eyes and black hair	$O_2 = 21$	$\pi_2 = \frac{1}{4}$	$E_2 = n\pi_2 = 100 \times \frac{1}{4} = 25$	$\frac{(O_2 - E_2)^2}{E_2} = \frac{(21-25)^2}{25} = 0.64$
Black eyes and brown hair	$O_3 = 14$	$\pi_3 = \frac{1}{4}$	$E_3 = n\pi_3 = 100 \times \frac{1}{4} = 25$	$\frac{(O_3 - E_3)^2}{E_3} = \frac{(14-25)^2}{25} = 4.84$
Black eyes and black hair	$O_4 = 34$	$\pi_4 = \frac{1}{4}$	$E_4 = n\pi_4 = 100 \times \frac{1}{4} = 25$	$\frac{(O_4 - E_4)^2}{E_4} = \frac{(34-25)^2}{25} = 3.24$
Total	$\sum_{i=1}^{4} O_i = 100$	$\sum_{i=1}^{4} \pi_i = 1$	$\sum_{i=1}^{4} E_i = 100 = \sum_{i=1}^{4} O_i$	$\chi^2_{cal} = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i} = 10.16$

1) Formulate the hypotheses for the test

$$H_0: \pi_1 = \frac{1}{4}, \pi_2 = \frac{1}{4}, \pi_3 = \frac{1}{4}, \pi_4 = \frac{1}{4}$$

(The people in this city have eye color and hair color proportions that do not differ from each other.)

$H_1:$ There is at least one π_i value that differs from the value specified in

H_0 ; $i = 1, 2, 3, 4$

(The people in this city have eye color and hair color proportions that are different from each other.)

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) The test statistic used is $\chi^2_{cal} = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$

$$= 1.44 + 0.64 + 4.84 + 3.24$$

$$= 10.16$$

4) The critical region (or statistics table values) is: Reject H_0 if $\chi^2_{cal} > \chi^2_{.05, df=4-1=3} = 7.81473$

5) Decision (accept H_0 or reject H_0): Reject H_0 and accept H_1

6) Conclusion: The people in this city have eye color and hair color proportions that are different at the 0.05 significance level.

2.2. Hypothesis Testing concerning the Median of a Single Population: Using the Wilcoxon Signed Method Ranks Test

1) Formulate the hypotheses for the test

H_0: $\text{Median} = M_0$

H_1: $\text{Median} \neq M_0$

Where $\text{Median}\ (M_0)$ is the median of the population (Population Median)

2) Specify the significance level of the test, equal to α

3) The test statistic is divided into two cases as follows:

3.1) **In the case where the sample size** $n \le 20$; the test statistic used is:

$$W = \sum_{i=1}^{n} R_i^{+}$$

Where R_i^{+} = the rank of observation $|D_i|$ unit i

(after ranking, assign a negative sign to the ranks whose D_i values are negative)

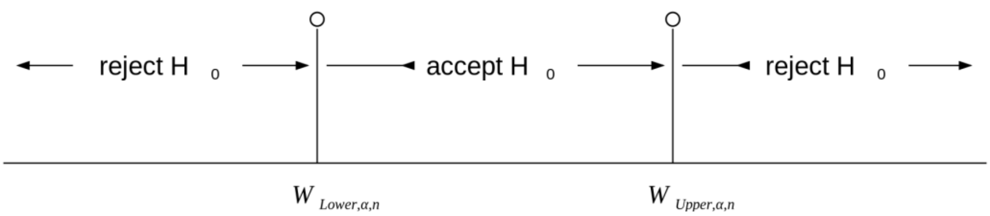
$D_i = x_i - \mathrm{M}_0$; M_0 is the median according to H_0

; $i = 1, 2, \ldots, n$

n = the total sample size, excluding the observations for which $D_i = 0$

<u>Note:</u> See the method for calculating W

<u>Decision Making</u>: Two-tailed test



- **<u>Reject</u>** H_0 and **<u>accept</u>** H_1 : if $W \le W_{\text{Lower},\alpha,n}$ or if $W \ge W_{\text{Upper},\alpha,n}$
- **<u>Accept</u>** H_0 : if $W_{\text{Lower},\alpha,n} < W < W_{\text{Upper},\alpha,n}$

where $W_{\text{Lower},\alpha,n}$ and $W_{\text{Upper},\alpha,n}$ are the two-tailed critical values obtained from the statistical table "Lower and Upper Critical Values W of Wilcoxon Signed-Ranks Test"

3.2) **In the case where the sample size** $n > 20$; the test statistic used is:

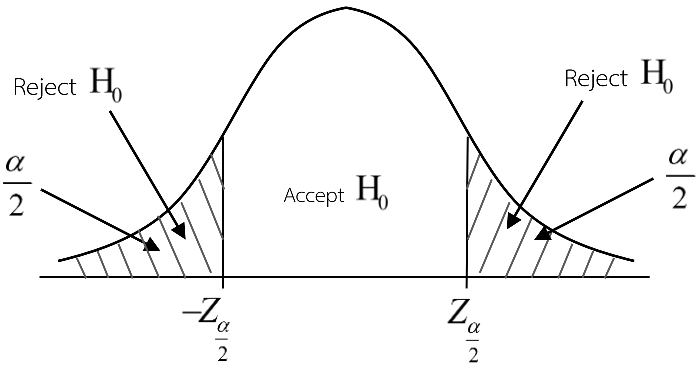
$$Z_{Cal} = \frac{W-\mu_W}{\sigma_W}$$

Where $W = \sum_{i=1}^{n} R_i^+$ (see the method for calculating W)

$$\mu_W = 0$$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}}$$

<u>Decision Making: Two-tailed test</u>



- **<u>Reject</u>** H_0 and **<u>accept</u>** H_1 : if $Z_{cal} < -Z_{\frac{\alpha}{2}}$ and if $Z_{cal} > Z_{\frac{\alpha}{2}}$
- **<u>accept</u>** H_0 : if $-Z_{\frac{\alpha}{2}} < Z_{cal} < Z_{\frac{\alpha}{2}}$

<u>Steps for calculating</u> W : <u>Case of testing the median of a single population</u>

1. Find the difference between the data value x_i and the median M_0 (M_0 is the value specified in H_0)

   Denote this difference by D_i

$$\boxed{D_i = x_i - M_0} \quad ; \qquad i = 1, 2, \ldots, n$$

2. Find the absolute value of D_i denoted by $|D_i|$
3. Arrange the $|D_i|$ values in ascending order and assign ranks to these values.
   - If any $|D_i|$ values are **<u>tied</u>**, use **<u>the method of average ranks</u>** **(do not assign ranks to observations for which $D_i = 0$)**

4. From the ranks obtained in step 3, **assign a negative sign** to the ranks corresponding to negative D_i values.
5. **Compute the total sum of the Positive signed ranks obtained; denote this** by W

Example 2.13 An instructor of a statistics course believes that the median score of this examination **does not differ from 50 points. Therefore**, the instructor randomly selected the exam scores of 13 students and obtained the following data:

Student number	exam score (x_i)	$D_i = x_i - M_0$ $= x_i - 50$	$\lvert D_i \rvert$	rank of $\lvert D_i \rvert$	R_i^+	R_i^-
1	57	$D_1 = 57-50 = 7$	7	4	4	
2	70	$D_2 = 70-50 = 20$	20	9	9	
3	42	$D_3 = 42-50 = -8$	8	5		-5
4	48	$D_4 = 48-50 = -2$	2	1		-1
5	77	$D_5 = 77-50 = 27$	27	11	11	
6	63	$D_6 = 63-50 = 13$	13	7	7	
7	45	$D_7 = 45-50 = -5$	5	2.5		-2.5
8	64	$D_8 = 64-50 = 14$	14	8	8	
9	55	$D_9 = 55-50 = 5$	5	2.5	2.5	
10	39	$D_{10} = 39-50 = -11$	11	6		-6
11	73	$D_{11} = 73-50 = 23$	23	10	10	
12	78	$D_{12} = 78-50 = 28$	28	12	12	
13	50	$D_{13} = 50-50 = 0$	0	-	-	-
					63.5	14.5

1. Test the instructor’s belief at the 0.05 significance level.

Solution

1) Formulate the hypotheses for the test

H_0 : $\text{Median} = 50$ exam scores

H_1 : $\text{Median} \neq 50$ exam scores

2) Specify the significance level of the test, equal to $\alpha = 0.05$
3) The test statistic used is the W-Test, **because n = 12 < 20**

$$W = \sum_{i=1}^{n} R_i^+ = 4+9+\ldots+12 = 63.5$$

4) Critical region, that is, reject H_0 if $W \leq W_{\text{Lower},0.05,12}$ or reject H_0 if $W \leq W_{\text{Upper},0.05,12}$

$W_{\text{Lower},0.05,12} =$………………13…………………..

$W_{\text{Upper},0.05,12} =$………………..65…………………

5) Decision: Accept H_0
6) Conclusion

2. If this instructor additionally selects a further sample of 10 students, making a total of 23 students, and asks for their exam scores, then calculates D_i, $|D_i|$ and the ranks of $|D_i|$ in the same way as in (1), the results are: the sum of the Positive Signed Ranks = 150 and the sum of the Negative Signed Ranks = −80. Based on the available data, we would like to know whether this instructor's belief is still correct. Show the method for performing the test at $\alpha = 0.05$ and state the conclusion obtained.

Solution

1) Formulate the hypotheses for the test

H_0 : $\text{Median} = 50$ exam scores

H_1 : $\text{Median} \neq 50$ exam scores

2) Specify the significance level of the test, equal to $\alpha = 0.05$
3) The test statistic used is the Z-Test, **because n = 12+10 = 22 > 20**

$$Z_{Cal} = \frac{W - \mu_W}{\sigma_W} = \frac{150-0}{61.6} = 2.43$$

Where $W = \sum_{i=1} R_i^+$

= (**<u>sum</u> of Positive Signed Ranks**)

$= 150$

$\mu_W = 0$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{22(22+1)(2(22)+1)}{6}} = 61.6$$

4) Critical region, that is, reject H_0 if $Z_{Cal} < -1.96$ or reject H_0 if $Z_{Cal} > 1.96$

since $Z_{cal} = 2.43 > 1.96$

5) Decision: Reject H_0

6) Conclusion: The median exam score in the statistics course is different from 50 points at $\alpha = 0.05$.

<u>Hypothesis Testing concerning the Distribution of a Population</u>

2.3. **<u>Hypothesis Testing concerning Two Related Populations: Using the Wilcoxon Signed Ranks Test with Paired Data</u>**

1) Formulate the hypotheses for the test

$H_0:$ The medians of the two populations are not different.

$H_1:$ The medians of the two populations are different.

2) Specify the significance level of the test, equal to α

3) The test statistic is divided into two cases as follows:

3.1) **In the case where the sample size** $n \le 20$; the test statistic used is

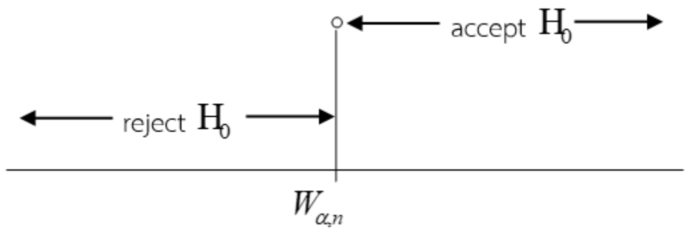
$$W = \text{Smaller}\{W^+, W^-\}$$

where $W^+ =$ the **<u>sum</u>** of the ranks of observations with **<u>positive</u>** D_i

W^{-} = the **sum** of the ranks of observations with **negative** D_i

Note: See the method for calculating W

Decision Making: Two-tailed test



- **Reject** H_0 and **accept** H_1 : if $W \le W_{\alpha,n}$
- **accept** H_0 : if $W > W_{\alpha,n}$

where $W_{\alpha,n}$ are the two-tailed critical values obtained from the statistical table "Critical Values for the Wilcoxon Signed-Ranks Test for n = 5 to 50"

3.2) **In the case where the sample size** $n > 20$; the test statistic used is

$$Z_{Cal} = \frac{W - \mu_W}{\sigma_W}$$

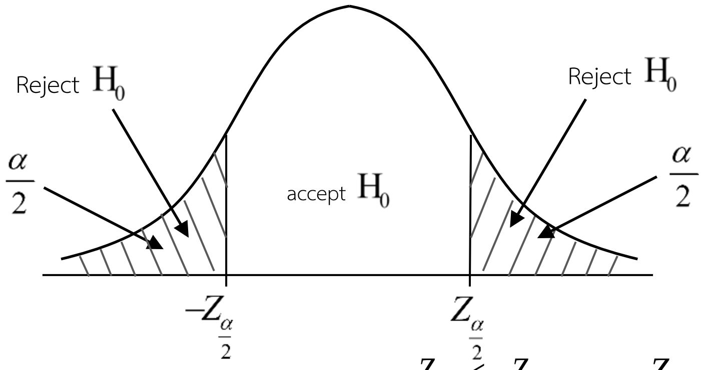
Where $W = \text{Smaller}\{W^{+}, W^{-}\}$ (see the method for calculating W)

$$\mu_W = \frac{n(n+1)}{4}$$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}}$$

n = the total sample size excluding the observations for which $D_i = 0$

Decision Making: Two-tailed test



• **<u>Reject</u>** H_0 and **<u>accept</u>** H_1 : if $Z_{cal} < -Z_{\frac{\alpha}{2}}$ or if $Z_{cal} > Z_{\frac{\alpha}{2}}$

• **<u>accept</u>** H_0: if $-Z_{\frac{\alpha}{2}} < Z_{cal} < Z_{\frac{\alpha}{2}}$

<u>Steps for calculating W : Case of testing two related populations</u>

1) Find the difference for each pair of observations; denote this difference by D_i

$$\boxed{D_i = x_i - y_i} \quad ; \qquad i = 1, 2, \ldots, n$$

where x_i is the observation in group 1 **(before)**.

y_i is the observation in group 2 **(after)**.

2) Find the absolute value of D_i denoted by $|D_i|$

3) Arrange the $|D_i|$ values in ascending order and assign ranks to these values.

If any $|D_i|$ values **<u>are tied</u>**, use the **<u>method of average ranks</u> (do not assign ranks to observations for which $D_i = 0$)**

4) **<u>Compute the sum</u>** of the ranks corresponding to **<u>positive</u>** $D_i = 0$ denoted by W^+

<u>Compute the sum</u> of the ranks corresponding to **<u>negative</u>** $D_i = 0$ denoted by W^-

5) Compute the value $W = \text{Smaller}\{\mathrm{W}^+, \mathrm{W}^-\}$

<u>Example 2.14</u> A sample of 7 first-year science students was selected. Their scores in Mathematics 1 and Physics in Everyday Life (each with a maximum score of 20 points) were collected as follows.

Student No.	Mathematics 1 (x_i)	Physics (y_i)	$D_i = x_i - y_i$	$\lvert D_i \rvert$	rank of $\lvert D_i \rvert$	W^+	W^-
1	10	6	$D_1 = 10 - 6 = 4$	4	2.5	2.5	
2	16	20	$D_2 = 16 - 20 = -4$	4	2.5		2.5
3	10	8	$D_3 = 10 - 8 = 2$	2	1	1	
4	4	12	$D_4 = 4 - 12 = -8$	8	5		5
5	2	8	$D_5 = 2 - 8 = -6$	6	4		4
6	14	4	$D_6 = 14 - 4 = 10$	10	6	6	
7	4	15	$D_7 = 4 - 15 = -11$	11	7		7
						$W^+ = 9.5$	$W^- = 18.5$

Determine whether the medians of the Mathematics 1 scores and the Physics in Everyday Life scores of this group of students are different at the 0.05 significance level.

Solution

1) Formulate the hypotheses for the test

$H_0:$ The medians of the Mathematics 1 scores and the Physics scores are <u>not different</u>.

$H_1:$ The medians of the Mathematics 1 scores and the Physics scores are <u>different</u>.

2) Specify the significance level of the test, equal to $\alpha = 0.05$
3) The test statistic used is the W-Test, because n = 7 ≤ 20

$W^+ = 2.5 + 1.0 + 6.0 = 9.5$
$W^- = 2.5 + 5.0 + 4.0 + 7.0 = 18.5$
$W = \min(W^+, W^-) = \min(9.5, 18.5) = 9.5$

$W^+ = 2.5 + 1.0 + 6.0 = 9.5$
$W^- = 2.5 + 5.0 + 4.0 + 7.0 = 18.5$
$W = \min(W^+, W^-) = \min(9.5, 18.5) = 9.5$

4) The critical region (or statistics table values) is:
Reject H_0 if $W \le W_L$ or $W \ge W_U$

Wilcoxon table (n=7, α=0.05, two-tail): W_L = 2, W_U = 26

Reject H_0 if $W \le W_L$ or $W \ge W_U$
From Wilcoxon Signed-Ranks table (n=7, α=0.05, two-tail): $W_L = 2$, $W_U = 26$

5) Decision (accept H_0 or reject H_0)
Accept H_0 (W = 9.5, between $W_L = 2$ and $W_U = 26$)

6) Conclusion
The medians of the Mathematics 1 scores and the Physics in Everyday Life
scores of this group of students are not different at the 0.05 significance level.

<u>Example 2.15</u> In an English 2 examination, a pretest was administered <u>before</u> the lesson content, and one week <u>after</u> the lesson the same test was given to the same group of students. From a sample of 26 students, the following data were obtained.

Student No.	Pretest (x_i)	Posttest (y_i)	$D_i = x_i - y_i$	$\lvert D_i \rvert$	rank $\lvert D_i \rvert$	W^+	W^-
1	46	76	-30	30	25		25
2	27	36	-9	9	7		7
3	37	53	-16	16	12.5		12.5
4	34	55	-21	21	18		18
5	20	12	8	8	6	6	
6	38	50	-12	12	10		10
7	10	36	-26	26	22		22
8	24	18	6	6	4	4	
9	20	21	-1	1	1		1
10	39	57	-18	18	15		15
11	16	27	-11	11	9		9
12	20	48	-28	28	23		23
13	47	70	-23	23	19		19
14	45	25	20	20	17	17	
15	40	50	-10	10	8		8
16	46	39	7	7	5	5	
17	32	51	-19	19	16		16
18	49	33	16	16	12.5	12.5	
19	45	69	-24	24	20		20
20	49	52	-3	3	2		2
21	44	60	-16	16	12.5		12.5
22	45	20	25	25	21	21	
23	16	12	4	4	3	3	
24	41	70	-29	29	24		24
25	48	64	-16	16	12.5		12.5
26	35	35	0	0		$W^+ = 68.5$	$W^- = 256.5$

We would like to know whether the medians of the pretest and posttest scores are different at the 0.10 significance level.

Solution

1) Formulate the hypotheses for the test

$H_0:$ The medians of the Pretest and Posttest scores are not different.

$H_1:$ The medians of the Pretest and Posttest scores are different.

2) Specify the significance level of the test, equal to $\alpha = 0.10$
3) The test statistic used is the Z-Test

because $n = 26 - 1 = 25 > 20$ (there is one difference value equal to $D_i = 0$,

therefore n s reduced)

$$Z_{Cal} = \frac{W - \mu_W}{\sigma_W} = \frac{68.5 - 162.5}{37.17} = -2.53$$

where $W = \text{Smaller}\{W^+, W^-\} = \text{Smaller}\{68.5, 256.5\} = 68.5$

$$\mu_W = \frac{n(n+1)}{4} = \frac{25(25+1)}{4} = 162.5$$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}} = \sqrt{\frac{25(25+1)(2(25)+1)}{24}} = 37.17$$

4) Critical region, that is, reject H_0 if $Z_{cal} < -1.645$ or reject H_0 if $Z_{cal} > 1.645$

since $Z_{cal} = -2.53 < -1.645$

5) Decision: Reject H_0 and accept H_1

6) Conclusion: The medians of the Pretest and Posttest scores are <u>different</u> at the 0.10 significance level.

2.4. Hypothesis Testing concerning Two Independent Populations: Using the Mann–Whitney U-Test

1) Formulate the hypotheses for the test

$H_0:$ The medians of the two populations are **<u>not different</u>**.

$H_1:$ The medians of the two populations are **<u>different</u>**.

2) Specify the significance level of the test, equal to α
3) The test statistic is divided into two cases as follows:

3.1) **In the case where the sample sizes are $n_1 \le 10$ and $n_2 \le 10$**:
The test statistic used is

$$U = \text{Smaller}\{U_1, U_2\}$$

Where $U_1 = \sum R_1 - \frac{n_1(n_1+1)}{2}$, $U_2 = \sum R_2 - \frac{n_2(n_2+1)}{2}$

$\sum R_1 =$ the **sum** of the ranks of the observations in group 1

$\sum R_2 =$ the **sum** of the ranks of the observations in group 2

Note: See the method for calculating U

Decision Making: Two-tailed test

- **Reject** H_0 and **accept** H_1 : if $U \le U_{\alpha, n_1, n_2}$
- **Accept** H_0 : if $U > U_{\alpha, n_1, n_2}$

Where U_{α, n_1, n_2} are the two-tailed critical values, obtained from the statistical table "Critical Values of U in the Mann-Whitney Test"

3.2) **In the case where the sample sizes are $n_1 > 10$** and $n_2 > 10$:
the test statistic used is

$$Z_{Cal} = \frac{U - \mu_U}{\sigma_U}$$

Where $U = \text{Smaller}\{U_1, U_2\}$: (see the method for calculating U)

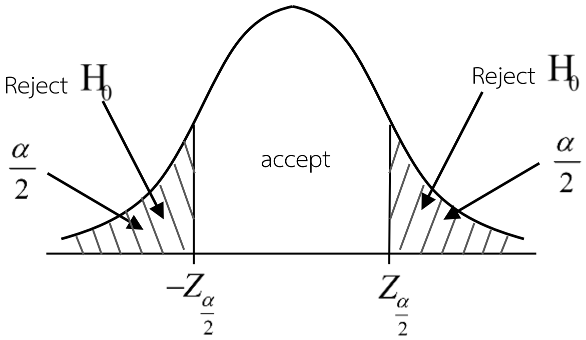
$$\mu_U = \frac{n_1 n_2}{2}$$

$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

$n_1 =$ is the sample size of group 1

$n_2 =$ is the sample size of group 2

Decision Making: Two-tailed test



- **Reject** H_0 and **accept** H_1 : if $Z_{cal} < -Z_{\frac{\alpha}{2}}$ or if $Z_{cal} > Z_{\frac{\alpha}{2}}$
- **Accept** H_0 : if $-Z_{\frac{\alpha}{2}} \le Z_{cal} \le Z_{\frac{\alpha}{2}}$

Steps for calculating U : Case of testing two independent populations

1. Arrange the data from both groups together in ascending order.
2. Assign ranks to the ordered data.
   - If any values are tied, use the method of average ranks.
3. **Compute the sum of the ranks for group 1**, denoted by $\sum R_1$
   Compute the sum of the ranks for group 2, denoted by $\sum R_2$
4. Calculate U_1, U_2 and U

Where $U = \text{Smaller}\{U_1, U_2\}$

$$U_1 = \sum R_1 - \frac{n_1(n_1+1)}{2}$$

$$U_2 = \sum R_2 - \frac{n_2(n_2+1)}{2}$$

and $\sum R_1 =$ **sum** of the ranks of the observations in group 1,

$\sum R_2 =$ **sum** of the ranks of the observations in group 2

$n_1 =$ sample size of group 1

$n_2 =$ sample size of group 2

Example 2.16 A study is conducted to compare the fuel consumption rate between 95-octane gasoline and 95-octane gasohol to determine whether they are different. The experiment is carried out using 14 new cars of the same model, which are divided equally into two groups. One group uses 95-octane gasoline, and the other group uses 95-octane gasohol. The cars are then driven on a test track, and the fuel consumption rate (measured in kilometers per liter) is recorded.

Gasoline 95 : A	13	11	10	9	7	9	8
Gasohol 95 : B	8	12	20	17	18	14	15

Show the method for performing the test using nonparametric statistics and state the conclusion at the 0.05 significance level.

Solution

Fuel consumption rate	rank	fuel type
7	1	A
8	2.5	A
8	2.5	B
9	4.5	A
9	4.5	A
10	6	A
11	7	A
12	8	B
13	9	A
14	10	B
15	11	B
17	12	B
18	13	B
20	14	B

1) Formulate the hypotheses for the test

H_0: The median fuel consumption rate of fuel A is <u>**not different**</u> from that of fuel B.

H_1: The median fuel consumption rate of fuel A is <u>**different**</u> from that of fuel B.

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) The test statistic used is $U = \text{Smaller}\{U_1, U_2\} = \text{Smaller}\{6.5, 42.5\} = 6.5$

where $U_1 = \sum R_1 - \frac{n_1(n_1+1)}{2} = 34.5 - \frac{7(7+1)}{2} = 6.5$

$$U_2 = \sum R_2 - \frac{n_2(n_2+1)}{2} = 70.5 - \frac{7(7+1)}{2} = 42.5$$

$\sum R_1 =$ the **sum** of the ranks of the observations in group 1 (fuel A)

$$= 1 + 2.5 + 4.5 + 4.5 + 6 + 7 + 9 = 34.5$$

$\sum R_2 =$ the **sum** of the ranks of the observations in group 2 (fuel B)

$$= 2.5 + 8 + 10 + 11 + 12 + 13 + 14 = 70.5$$

4) Critical region, that is, reject H_0 if $U \le U_{\alpha, n_1, n_2}$

$U_{\alpha, n_1, n_2} =$....................8.............................

5) Decision

Reject H_0 ($U = 6.5 \le U_{critical} = 8$)

Reject H_0 ($U = 6.5 \le U_{critical} = 8$)

6) Conclusion

The median fuel consumption rate of Gasoline 95 and Gasohol 95 are different
at the 0.05 significance level.
(Gasohol 95 gives higher fuel efficiency than Gasoline 95)
The median fuel consumption rate of Gasoline 95 and Gasohol 95 are different
at the 0.05 significance level.
(Gasohol 95 gives higher fuel efficiency (km/litre) than Gasoline 95)

Example 2.17 An academic advisor wishes to compare the academic performance of 25 advisees, consisting of 12 male students and 13 female students. The advisor collects their GPA data as follows:

male students (group 1)	2.72	3.10	3.75	1.92	2.41	3.72	2.35	3.80	3.25	2.71	3.85	3.90	
female students (group 2)	3.10	2.10	2.62	1.62	2.85	1.95	3.28	1.50	3.71	2.37	3.86	3.88	3.89

We would like to know whether the median academic performance of male students differs from that of female students at the 0.05 significance level.

Solution

Take the data in the above table and assign ranks by arranging the GPAs from smallest to largest and separating them by group as follows.

GPA	rank	group
1.50	1	2
1.62	2	2
1.92	3	1
1.95	4	2
2.10	5	2
2.35	6	1
2.37	7	2
2.41	8	1
2.62	9	2
2.71	10	1
2.72	11	1
2.85	12	2
3.10	13.5	1
3.10	13.5	2
3.25	15	1
3.28	16	2
3.71	17	2

GPA	rank	group
3.72	18	1
3.75	19	1
3.80	20	1
3.85	21	1
3.86	22	2
3.88	23.5	2
3.89	23.5	2
3.90	25	1

1) Formulate the hypotheses for the test

$H_0:$ The median of the academic performance of male students is **<u>not different</u>** from that of female students.

$H_1:$ The median of the academic performance of male students is **<u>different</u>** from that of female students.

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) The test statistic used is the Z-test because $n_1 = 12 > 10$ and $n_2 = 13 > 10$

$$Z_{Cal} = \frac{U-\mu_U}{\sigma_U} = \frac{64.5-78}{18.38} = -0.73$$

Where $U = \text{Smaller}\{U_1, U_2\} = \text{Smaller}\{91.5, 64.5\} = 64.5$

$$U_1 = \sum R_1 - \frac{n_1(n_1+1)}{2} = 169.5 - \frac{12(12+1)}{2} = 91.5$$

$$U_2 = \sum R_2 - \frac{n_2(n_2+1)}{2} = 155.5 - \frac{13(13+1)}{2} = 64.5$$

$\sum R_1 =$ The **<u>sum</u>** of ranks of data in Group 1 (male)

$$= 3+6+8+10+11+13.5+15+18+19+20+21+25 = 169.5$$

$\sum R_2 =$ The **<u>sum</u>** of ranks of data in Group 2 (female)

$$= 1+2+4+5+7+9+12+13.5+16+17+22+23+24 = 155.5$$

$$\mu_U = \frac{n_1 n_2}{2} = \frac{12\times 13}{2} = 78$$

$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1+n_2+1)}{12}} = \sqrt{\frac{(12\times 13)(12+13+1)}{12}} = 18.38$$

4) Critical region, that is, we will reject H_0 if $Z_{cal} < -1.96$ or reject H_0 if $Z_{cal} > 1.96$

Because $-1.96 < Z_{cal} = -0.73 < 1.96$

5) Decision: Accept H_0
6) Conclusion: The median of the academic performance of male students is not different from that of female students at the significance level of 0.05

2.5. Hypothesis testing for populations of 3 or more groups ($k \geq 3$) that are independent of one another: by the Kruskal–Wallis H-Test

1) Formulate the hypotheses for the test

$H_0: \ M_1 = M_2 = M_3 = \ldots = M_k$

$H_1:$ There exists at least one population median that is different from the others.

Where $M_1, M_2, M_3, \ldots, M_k$ is the median in the population of group $1, 2, 3, \ldots, k$ respectively.

$k =$ number of groups ; $k \geq 3$

2) Specify the significance level of the test, equal to α
3) The test statistic used is

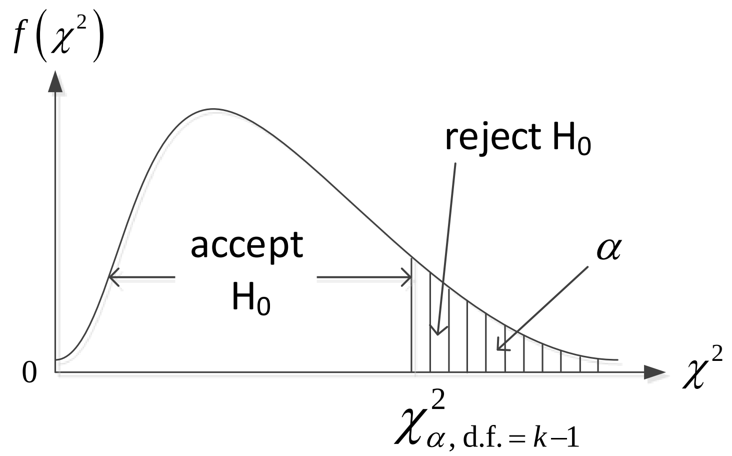
$$H = \left[\frac{12}{n(n+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} \right] - 3(n+1)$$

Where $R_j =$ the sum of ranks of the data in group j; $j = 1, 2, \ldots, k$

$n_j =$ the sample size in group j

$n = n_1 + n_2 + \ldots + n_k$

4) Decision



- <u>Reject</u> H_0 and <u>accept</u> H_1 if $H > \chi^2_{\alpha,\text{d.f.}=k-1}$
- <u>Accept</u> H_0 if $H \le \chi^2_{\alpha,\text{d.f.}=k-1}$

<u>Steps for calculating the test statistic H: in the case of testing populations of 3 or more independent groups</u>

1. Arrange the data from all k groups together, from the smallest to the largest, $k \ge 3$
2. **<u>Assign ranks</u>** to the ordered data.
   • If the data contain **<u>tied values</u>**, use the method of assigning **<u>the average rank.</u>**
3. - **<u>Find the sum of ranks of Group 1</u>**, denoted by R_1
   - **<u>Find the sum of ranks of Group 2</u>**, denoted by R_2
   - **<u>Find the sum of ranks of Group k</u>**, denoted by R_k
4. Compute the value of H according to the H formula

<u>Example 2.18</u> In a study to compare training methods on basic safety in factory buildings, the manager is interested in 4 training methods as follows:

<u>Method 1</u> Learning by using a manual

<u>Method 2</u> Lecture by an instructor

<u>Method 3</u> Watching a video

<u>Method 4</u> Group discussion

A random sample of 20 workers in the factory was selected. Each worker was randomly assigned to receive one of the four training methods. The training lasted for 3 days. After that, a test was administered to measure the knowledge of the workers who received each of the 4 training methods. The full score on the test is 100 points. The data obtained are as follows:

Method 1	Method 2	Method 3	Method 4
59 (4)	57 (3)	55 (2)	54 (1)
60 (5)	61 (6)	65 (7)	66 (8)
69 (9)	70 (10)	71 (11)	72 (12)
78 (16)	73 (13)	75 (14)	77 (15)
85 (20)	82 (19)	81 (18)	80 (17)
$R_1 = 54$	$R_2 = 51$	$R_3 = 52$	$R_4 = 53$

Determine whether the 4 training methods yield different results. State the conclusion at the 0.01 significance level.

Solution

1) Formulate the hypotheses for the test

$H_0:\ M_1 = M_2 = M_3 = M_4$ (The 4 training methods do not yield different results)

$H_1:$ At least one population median is different from the others.

2) Specify the significance level of the test, equal to $\alpha = 0.01$

3) The test statistic used is

$$\text{H} = \left[\frac{12}{n(n+1)}\sum_{j=1}^{4}\frac{R_j^2}{n_j}\right] - 3(n+1)$$

$$= \left[\frac{12}{n(n+1)}\left(\frac{R_1^2}{n_1}+\frac{R_2^2}{n_2}+\frac{R_3^2}{n_3}+\frac{R_4^2}{n_4}\right)\right] - 3(n+1)$$

$$= \left[\frac{12}{20(20+1)}\left(\frac{(54)^2}{5}+\frac{(51)^2}{5}+\frac{(52)^2}{5}+\frac{(53)^2}{5}\right)\right] - 3(20+1)$$

$$\text{H} = \left[\frac{12}{20(21)}\times\left(\frac{11,030}{5}\right)\right] - 63 = 0.02857$$

4) Critical region, that is reject H_0 if $\text{H} > 11.3449$ and accept H_0 **if** $\text{H} \le 11.3449$

because $\mathrm{H} = 0.02857 < 11.3449$

5) Decision: Accept H_0
6) Conclusion: The 4 training methods do not yield different results at the 0.01 significance level.

Example 2.19 In the production of cylindrical metal rods using 3 automatic machines, when a random sample of 5 rods from each machine is taken, the diameters of each rod measured are as follows (unit: millimeters)

Machine No.	1	52	53	50	49	51
	2	45	38	48	43	35
	3	31	42	30	34	32

Test whether the medians of the diameters of the cylindrical metal rods from each machine are different or not at the 0.05 significance level.

Solution

Machine No. 1	Machine No. 2	Machine No. 3
49 (11)	35 (5)	30 (1)
50 (12)	38 (6)	31 (2)
51 (13)	43 (8)	32 (3)
52 (14)	45 (9)	34 (4)
53 (15)	48 (10)	42 (7)
$R_1 = 65$	$R_2 = 38$	$R_3 = 17$

1) Formulate the hypotheses for the test

$H_0:\ M_1 = M_2 = M_3$ (The medians of the diameters of the cylindrical metal rods from all 3 machines are not different.)

$H_1:$ There exists at least one median of the diameters of the cylindrical metal rods from one machine that is different from the others.

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) The test statistic used is $\mathrm{H} = \left[\frac{12}{n(n+1)}\sum_{j=1}^{3}\frac{R_j^2}{n_j}\right] - 3(n+1)$

$$= \left[\frac{12}{n(n+1)}\left(\frac{R_1^2}{n_1}+\frac{R_2^2}{n_2}+\frac{R_3^2}{n_3}\right)\right] - 3(n+1)$$

$$= \left[\frac{12}{15(15+1)}\left(\frac{(65)^2}{5}+\frac{(38)^2}{5}+\frac{(17)^2}{5}\right)\right] - 3(15+1)$$

$$\mathrm{H} = \left[\frac{12}{15(16)}\times\left(\frac{5,958}{5}\right)\right] - 48 = 11.58$$

4) Critical region, that is reject H_0 if $\mathrm{H} > 5.9915$ and accept H_0 **if** $\mathrm{H} \le 5.9915$

Because $\mathrm{H} = 11.58 < 5.9915$

5) Decision: Reject: Reject H_0

6) Conclusion: The medians of the diameters of the cylindrical metal rods from all 3 machines are different at the 0.05 significance level.

<u>Multiple Comparison</u>

It is a method of comparing the population medians pairwise by testing hypotheses to determine which population group causes the rejection of H_0 in the H-Test. In this context, multiple comparisons are performed using Dunn's Procedure.

• **<u>Dunn's Procedure</u>**

It is a method of comparing the population medians pairwise to determine whether they are different, by considering **<u>the absolute value of the difference between the average ranks</u>** for **<u>each pair</u>** and the Critical Range. Then, observe which pairs have values exceeding this specified Critical Range, and conclude that those population pairs have significantly different medians.

• **<u>Steps for testing hypotheses concerning population medians pairwise by Dunn's Procedure</u>**

The number of population median pairs that must be tested is

$$= {}^{k}C_2 = \binom{k}{2} = \frac{k!}{2!(k-2)!}$$

Where k is the number of population groups

Formulate the hypotheses for the test

$H_0 : M_i = M_j \quad ; \quad i \neq j$

$H_1 : M_i \neq M_j$

Specify the significance level of the test, equal to α

Compute the Critical Range $= Z_{\frac{\alpha}{2m}} \sqrt{\frac{n(n+1)}{12}\left(\frac{1}{n_i}+\frac{1}{n_j}\right)}$

Where $m = \frac{k(\mathrm{k}-1)}{2} \quad ; \quad n = n_1 + n_2 + \ldots + n_k$

n_i and n_j are the sample sizes of groups i and j, respectively.

Decision Compare $\left|\overline{R}_i - \overline{R}_j\right|$ with the Critical Range.

- <u>Reject</u> H_0 and <u>accept</u> H_1 if $\left|\overline{R}_i - \overline{R}_j\right| >$ Critical Range and conclude that the population medians of that pair are **<u>different</u>**.
- <u>Accept</u> H_0 if $\left|\overline{R}_i - \overline{R}_j\right| \leq$ Critical Range and conclude that the population medians of that pair are **<u>not different</u>**.

<u>Example 2.20</u> Use Dunn's Procedure to determine which pairs of medians of the diameters of the cylindrical metal rods are different (perform pairwise comparison of medians for 3 pairs).

Solution

In this case, pairwise comparison of medians will be performed for 3 pairs as follows:

I. <u>Comparison of the population medians of Groups 1 and 2</u>

1) Formulate the hypotheses for the test

$H_0 : M_1 = M_2$ (The medians of the diameters of the cylindrical metal rods from Machines 1 and 2 are not different.)

$H_1:\ M_1 \neq M_2$ (The medians of the diameters of the cylindrical metal rods from Machines 1 and 2 are different.)

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) Compute the Critical Range $= Z_{\frac{\alpha}{2m}}\sqrt{\frac{n(n+1)}{12}\left(\frac{1}{n_i}+\frac{1}{n_j}\right)}$

$$= Z_{\frac{\alpha}{2m}}\sqrt{\frac{n(n+1)}{12}\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$$

$$= Z_{\frac{0.05}{2(3)}}\sqrt{\frac{15(15+1)}{12}\left(\frac{1}{5}+\frac{1}{5}\right)}$$

$$= 2.326(2.83)$$

Therefore, Critical Range $= 6.58258$

Where $m = \frac{k(\mathrm{k}-1)}{2} = \frac{3(3-1)}{2} = 3$

Compare $\left|\overline{R}_1 - \overline{R}_2\right|$ with the Critical Range

The average rank of Group 1 is $\overline{R}_1 = \frac{R_1}{n_1} = \frac{65}{5} = 13$

The average rank of Group 2 is $\overline{R}_2 = \frac{R_2}{n_2} = \frac{38}{5} = 7.6$

The average rank of Group 3 is $\overline{R}_3 = \frac{R_3}{n_3} = \frac{17}{5} = 3.4$

Therefore, $\left|\overline{R}_1 - \overline{R}_2\right| = |13-7.6| = |5.4| <$ Critical Range $= 6.58$

4) Decision: Accept H_0
5) Conclusion: The medians of the diameters of the cylindrical metal rods from Machines 1 and 2 are not different at the 0.05 significance level.

II. **Comparison of the population medians of Groups 1 and 3**

1) Formulate the hypotheses for the test

$H_0: M_1 = M_3$ (The medians of the diameters of the cylindrical metal rods from Machines 1 and 3 are not different.)

$H_1: M_1 \neq M_3$ (The medians of the diameters of the cylindrical metal rods from Machines 1 and 3 are different.)

2) Specify the significance level of the test, equal to $\alpha = 0.05$

Compare $|\overline{R}_1 - \overline{R}_3|$ with the Critical Range

Thus $|\overline{R}_1 - \overline{R}_3| = |13 - 3.4| = |9.6| = 9.6 >$ Critical Range $= 6.58$

3) Decision: Reject H_0

4) Conclusion: The medians of the diameters of the cylindrical metal rods from Machines 1 and 3 are different at the 0.05 significance level.

III. **<u>Comparison of the population medians of Groups 2 and 3</u>**

1) Formulate the hypotheses for the test

$H_0: M_2 = M_3$ (The medians of the diameters of the cylindrical metal rods from Machines 2 and 3 are <u>not different</u>.)

$H_1: M_2 \neq M_3$ (The medians of the diameters of the cylindrical metal rods from Machines 2 and 3 are different.)

2) Specify the significance level of the test, equal to $\alpha = 0.05$

Compare $|\overline{R}_2 - \overline{R}_3|$ with the Critical Range

Thus $|\overline{R}_2 - \overline{R}_3| = |7.6 - 3.4| = |4.2| = 4.2 <$ Critical Range $= 6.58$

3) Decision: Accept H_0

4) Conclusion: The medians of the diameters of the cylindrical metal rods from Machines 2 and 3 are not different at the 0.05 significance level.

<u>Overall Conclusion</u>

• Arrange the average ranks from the smallest to the largest, together with the group labels.

• From the results of Tests I, II, and III, if any pair of medians is not different, draw a straight line connecting those groups, as follows:

$$\overline{R}_3 = 3.4 \qquad \overline{R}_2 = 7.6 \qquad \overline{R}_1 = 13$$

<u>The results can be concluded as follows:</u>

1) Machines 1 and 3 have <u>different</u> medians of the diameters.
2) Machine 2 has a median diameter that is <u>not different</u> from those of Machines 1 and 3, respectively.

2.6. Hypothesis testing concerning the rank correlation coefficient of two variables: by Spearman's Test for Rank Correlation

<u>Rank Correlation Coefficient</u>

Refers to the value used to measure both the magnitude and the direction of the relationship between two variables, by assigning ranks to the data in both groups first and then calculating the rank correlation coefficient.

Let r_s denote the rank correlation coefficient of the sample data by Spearman's method, also called "Spearman's Test for Rank Correlation"

$$r_s = 1 - \frac{6\sum_{i=1}^{n} D_i^2}{n(n^2-1)}$$

Where $-1 \le r_s \le +1$

D_i = the rank of the x_i data - the rank of the y_i data

(D_i = the difference in ranks of variables x and y for pair i)

n = the sample size

<u>Steps for hypothesis testing concerning the rank correlation coefficient</u>

1) Formulate the hypotheses for the test

$H_0:\ \rho_S = 0$ (The variables x and y are **<u>not</u>** related.)

$H_1:\ \rho_S \neq 0$ (The variables x and y are related.)

Where ρ_S is the rank correlation coefficient in the population data by Spearman's method.

2) Specify the significance level of the test, equal to equal to α

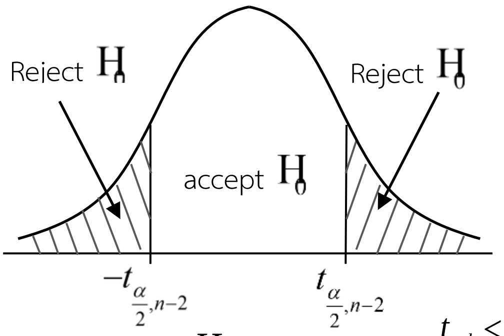
3) The test statistic, divided into 2 cases as follows:

3.1) **In the case where the sample size is** $n \leq 30$; the test statistic is:

$$t_{cal} = \frac{r_S - \mu_{r_S}}{S_{r_S}} = \frac{r_S - 0}{\sqrt{\frac{1-r_S^2}{n-2}}} = r_S\sqrt{\frac{n-2}{1-r_S^2}}$$

<u>Note</u>: See the method of calculating r_S

<u>Decision</u>: For the two-tailed test



- <u>Reject</u> H_0 and <u>accept</u> H_1 : if $t_{cal} < -t_{\frac{\alpha}{2},n-2}$ or if $t_{cal} > t_{\frac{\alpha}{2},n-2}$
- <u>accept</u> H_0 : if $-t_{\frac{\alpha}{2},n-2} \leq t_{cal} \leq t_{\frac{\alpha}{2},n-2}$

Where $t_{\frac{\alpha}{2},n-2}$ is the statistics table value t that has the right-tail area equal to $\frac{\alpha}{2}$ and has $\text{d.f.} = n-2$

3.2) **In the case where the sample size is** $n > 30$; the test statistic is:

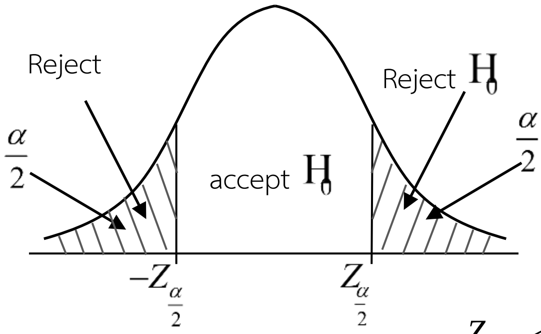
$$Z_{cal} = \frac{r_S - \mu_{r_S}}{\sigma_{r_S}}$$

Where $r_s = 1 - \frac{6\sum_{i=1}^{n} D_i^2}{n(n^2-1)}$ **;**

$$\mu_{r_s} = 0$$

$$\sigma_{r_s} = \sqrt{\frac{1}{n-1}}$$

<u>Decision</u>: For the two-tailed test



- <u>Reject</u> H_0 and <u>accept</u> H_1 : if $Z_{cal} < -Z_{\frac{\alpha}{2}}$ or if $Z_{cal} > Z_{\frac{\alpha}{2}}$
- <u>Accept</u> H_0 : if $-Z_{\frac{\alpha}{2}} \leq Z_{cal} \leq Z_{\frac{\alpha}{2}}$

<u>Steps for calculating</u> r_s

1. Assign ranks to the data for variables x and y **<u>separately in each group.</u>**
   - If the data contain <u>tied values</u>, use the method of assigning <u>the average rank.</u>

2. Find the difference in ranks of variables x and y for pair i denoted by D_i

D_i = The rank of the data x_i - The rank of the data y_i

; $i = 1, 2, \ldots, n$

3. Compute the value of r_s using the following formula:

$$r_s = 1 - \frac{6\sum_{i=1}^{n} D_i^2}{n(n^2-1)}$$

n = the sample size

Example 2.21 In a study of the relationship between mid-term examination scores and final examination scores in the course "Statistics in Everyday Life," the researcher randomly selected 10 students enrolled in this course and obtained their mid-term and final

Person No. (i)	Mid-term score (x_i)	Final score (y_i)	Rank of x_i	Rank of y_i	D_i	D_i^2
1	38	82	7	8	-1	1
2	28	65	3	2	1	1
3	35	69	5	4	1	1
4	40	74	8	6	2	4
5	36	72	6	5	1	1
6	27	51	2	1	1	1
7	45	85	9.5	9	0.5	0.25
8	25	68	1	3	-2	4
9	29	75	4	7	-3	9
10	45	89	9.5	10	-0.5	0.25

examination scores as follows:

Compute the value of r_s and explain the meaning of the value obtained.

Solution

Where

$$r_s = 1-\frac{6\sum_{i=1}^{n} D_i^2}{n(n^2-1)}$$

$$= 1-\frac{6\sum_{i=1}^{10} D_i^2}{10(10^2-1)}$$

$$= 1-\left(\frac{6\times 22.5}{10(10^2-1)}\right)$$

$$= 1-0.1364 \quad = \ 0.8636$$

<u>Explain the meaning of the value</u> r_s

r_s = 0.8636 indicates a strong positive relationship between mid-term and final examination scores. Students who achieve higher scores on the mid-term tend to achieve higher scores on the final examination as well, and vice versa.

r_s = 0.8636 indicates a strong positive relationship between mid-term and final examination scores. Students who achieve higher scores on the mid-term tend to achieve higher scores on the final examination as well, and vice versa.

<u>Example 2.22</u> In ranking the level of interest of news articles from 5 newspaper brands by two well-known radio news readers, the data obtained are shown in the table. Compute the value of r_s and test whether the two news readers have related opinions at $\alpha = 0.05$

	Newspaper					
Newscaster	Thai Rath	Daily News	Khao Sod	Kom Chad Luek	Krungthep Turakij	
Person No. 1 (x_i)	2	3	4	1	5	
Person No. 2 (y_i)	5	2	1	3	4	
D_i (Rank of x_i - Rank of y_i)	-3	1	3	-2	1	
D_i^2	9	1	9	4	1	$\sum D_i^2 = 24$

Solution

1) Formulate the hypotheses for the test

$H_0:\ \rho_S = 0$ (The two news readers <u>do not have related</u> opinions.)

$H_1:\ \rho_S \neq 0$ (The two news readers <u>have related</u> opinions.)

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) The test statistic used is $t_{cal} = r_S\sqrt{\frac{n-2}{1-r_S^2}}$ (Since $n = 5 < 30$ the t-test is used.)

$$= (-0.2)\sqrt{\frac{5-2}{1-(-0.2)^2}}$$

$$= -0.35$$

Where $r_s = 1-\frac{6\sum_{i=1}^{n} D_i^2}{n(n^2-1)}$

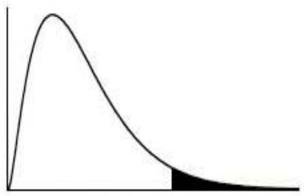
$$= 1-\left(\frac{6\times 24}{5(5^2-1)}\right)$$

$$= 1-1.2 \quad = -0.2$$

4) Critical region, that is reject H_0 if $t_{cal} < -3.182$ or reject H_0 if $t_{cal} > 3.182$

Because $-3.182 \le t_{cal} = -0.35 \le 3.182$

5) Decision: Accept H_0
6) Conclusion: The two news readers do not have related opinions at the 0.05 significance level.



Percentage points of the χ^2 distributions

d.f.	$\chi^2_{0.995}$	$\chi^2_{0.990}$	$\chi^2_{0.975}$	$\chi^2_{0.950}$	$\chi^2_{0.900}$	$\chi^2_{0.100}$	$\chi^2_{0.050}$	$\chi^2_{0.025}$	$\chi^2_{0.010}$	$\chi^2_{0.005}$	d.f.
1	0.0000393	0.0001571	0.0009821	0.0039321	0.0157908	2.70554	3.84146	5.02389	6.63490	7.87944	1
2	0.0100251	0.0201007	0.0506356	0.102587	0.210721	4.60517	5.99146	7.37776	9.21034	10.5966	2
3	0.0717218	0.114832	0.215795	0.351846	0.584374	6.25139	7.81473	9.34840	11.3449	12.8382	3
4	0.206989	0.297109	0.484419	0.710723	1.063623	7.77944	9.48773	11.1433	13.2767	14.8603	4
5	0.411742	0.554298	0.831212	1.145476	1.61031	9.23636	11.0705	12.8325	15.0863	16.7496	5
6	0.675727	0.872090	1.237344	1.63538	2.20413	10.6446	12.5916	14.4494	16.8119	18.5476	6
7	0.989256	1.239042	1.68987	2.16735	2.83311	12.0170	14.0671	16.0128	18.4753	20.2777	7
8	1.344413	1.646497	2.17973	2.73264	3.48954	13.3616	15.5073	17.5345	20.0902	21.9550	8
9	1.734933	2.087901	2.70039	3.32511	4.16816	14.6837	16.9190	19.0228	21.6660	23.5894	9
10	2.15586	2.55821	3.24697	3.94030	4.86518	15.9872	18.3070	20.4832	23.2093	25.1882	10
11	2.60322	3.05348	3.81575	4.57481	5.57778	17.2750	19.6751	21.9200	24.7250	26.7568	11
12	3.07382	3.57057	4.40379	5.22603	6.30380	18.5493	21.0261	23.3367	26.2170	28.2995	12
13	3.56503	4.10692	5.00875	5.89186	7.04150	19.8119	22.3620	24.7356	27.6882	29.8195	13
14	4.07467	4.66043	5.62873	6.57063	7.78953	21.0641	23.6848	26.1189	29.1412	31.3193	14
15	4.60092	5.22935	6.26214	7.26094	8.54676	22.3071	24.9958	27.4884	30.5779	32.8013	15
16	5.14221	5.81221	6.90766	7.96165	9.31224	23.5418	26.2962	28.8454	31.9999	34.2672	16
17	5.69722	6.40776	7.56419	8.67176	10.0852	24.7690	27.5871	30.1910	33.4087	35.7185	17
18	6.26480	7.01491	8.23075	9.39046	10.8649	25.9894	28.8693	31.5264	34.8053	37.1565	18
19	6.84397	7.63273	8.90652	10.1170	11.6509	27.2036	30.1435	32.8523	36.1909	38.5823	19
20	7.43384	8.26040	9.59078	10.8508	12.4426	28.4120	31.4104	34.1696	37.5662	39.9968	20
21	8.03365	8.89720	10.28290	11.5913	13.2396	29.6151	32.6706	35.4789	38.9322	41.4011	21
22	8.64272	9.54249	10.9823	12.3380	14.0415	30.8133	33.9244	36.7807	40.2894	42.7957	22
23	9.26042	10.19572	11.6886	13.0905	14.8480	32.0069	35.1725	38.0756	41.6384	44.1813	23
24	9.88623	10.8564	12.4012	13.8484	15.6587	33.1962	36.4150	39.3641	42.9798	45.5585	24
25	10.5197	11.5240	13.1197	14.6114	16.4734	34.3816	37.6525	40.6465	44.3141	46.9279	25
26	11.1602	12.1981	13.8439	15.3792	17.2919	35.5632	38.8851	41.9232	45.6417	48.2899	26
27	11.8076	12.8785	14.5734	16.1514	18.1139	36.7412	40.1133	43.1945	46.9629	49.6449	27
28	12.4613	13.5647	15.3079	16.9279	18.9392	37.9159	41.3371	44.4608	48.2782	50.9934	28
29	13.1211	14.2565	16.0471	17.7084	19.7677	39.0875	42.5570	45.7223	49.5879	52.3356	29
30	13.7867	14.9535	16.7908	18.4927	20.5992	40.2560	43.7730	46.9792	50.8922	53.6720	30
40	20.7065	22.1643	24.4330	26.5093	29.0505	51.8051	55.7585	59.3417	63.6907	66.7660	40
50	27.9907	29.7067	32.3574	34.7643	37.6886	63.1671	67.5048	71.4202	76.1539	79.4900	50
60	35.5345	37.4849	40.4817	43.1880	46.4589	74.3970	79.0819	83.2977	88.3794	91.9517	60
70	43.2752	45.4417	48.7576	51.7393	55.3289	85.5270	90.5312	95.0232	100.425	104.215	70
80	51.1719	53.5401	57.1532	60.3915	64.2778	96.5782	101.879	106.629	112.329	116.321	80
90	59.1963	61.7541	65.6466	69.1260	73.2911	107.565	113.145	118.136	124.116	128.299	90
100	67.3276	70.0649	74.2219	77.9295	82.3581	118.498	124.342	129.561	135.807	140.169	100

Lower and Upper Critical Values W of Wilcoxon Signed-Ranks Test

ONE-TAIL	$\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.01$	$\alpha = 0.005$
TWO-TAIL	$\alpha = 0.1$	$\alpha = 0.05$	$\alpha = 0.02$	$\alpha = 0.01$
n		*(Lower, Upper)*		
5	0,15	—,—	—,—	—,—
6	2,19	0,21	—,—	—,—
7	3,25	2,26	0,28	—,—
8	5,31	3,33	1,35	0,36
9	8,37	5,40	3,42	1,44
10	10,45	8,47	5,50	3,52
11	13,53	10,56	7,59	5,61
12	17,61	13,65	10,68	7,71
13	21,70	17,74	12,79	10,81
14	25,80	21,84	16,89	13,92
15	30,90	25,95	19,101	16,104
16	35,101	29,107	23,113	19,117
17	41,112	34,119	27,126	23,130
18	47,124	40,131	32,139	27,144
19	53,137	46,144	37,153	32,158
20	60,150	52,158	43,167	37,173

Source: Adapted from Table 2 of F. Wilcoxon and R. A. Wilcox, *Some Rapid Approximate Statistical Procedures*
(Pearl River, NY: Lederle Laboratories, 1964), with permission of the American Cyanamid Company.

Critical Values for the Wilcoxon Signed-Ranks Test for n = 5 to 50

	One Tail Probability					One Tail Probability			
n	0.05	0.025	0.01	0.005	n	0.05	0.025	0.01	0.005
	Two Tail Probability					Two Tail Probability			
n	0.10	0.05	0.02	0.01	n	0.10	0.05	0.02	0.01
5	1				28	130	117	102	92
6	2	1			29	141	127	111	100
7	4	2	0		30	152	137	120	109
8	6	4	2	0	31	163	148	130	118
9	8	6	3	2	32	175	159	141	128
10	11	8	5	3	33	188	171	151	138
11	14	11	7	5	34	201	183	162	149
12	17	14	10	7	35	214	195	174	160
13	21	17	13	10	36	228	208	186	171
14	26	21	16	13	37	242	222	198	183
15	30	25	20	16	38	256	235	211	195
16	36	30	24	19	39	271	250	224	208
17	41	35	28	23	40	287	264	238	221
18	47	40	33	28	41	303	279	252	234
19	54	46	38	32	42	319	295	267	248
20	60	52	43	37	43	336	311	281	262
21	68	59	49	43	44	353	327	297	277
22	75	66	56	49	45	371	344	313	292
23	83	73	62	55	46	389	361	329	307
24	92	81	69	61	47	408	379	345	323
25	101	90	77	68	48	427	397	362	339
26	110	98	85	76	49	446	415	380	356
27	120	107	93	84	50	466	434	398	373

Critical Values of U in the Mann - Whitney Test

Critical Values of U at α = 0.025 with direction predicted or at α = 0.05 with direction not predicted.

n_2 \ n_1	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2	0	0	0	0	0	0	0	1	1	1	1	1	2	2	2	2
3	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8
4	1	2	3	4	4	5	6	7	8	9	10	11	11	12	13	13
5	2	3	5	6	7	8	9	11	12	13	14	15	17	18	19	20
6	3	5	6	8	10	11	13	14	16	17	19	21	22	24	25	27
7	5	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34
8	6	8	10	13	15	17	19	22	24	26	29	31	34	36	38	41
9	7	10	12	15	17	20	23	26	28	31	34	37	39	42	45	48
10	8	11	14	17	20	23	26	29	33	36	39	42	45	48	52	55
11	9	13	16	19	23	26	30	33	37	40	44	47	51	55	58	62
12	11	14	18	22	26	29	33	37	41	45	49	53	57	61	65	69
13	12	16	20	24	28	33	37	41	45	50	54	59	63	67	72	76
14	13	17	22	26	31	36	40	45	50	55	59	64	67	74	78	83
15	14	19	24	29	34	39	44	49	54	59	64	70	75	80	85	90
16	15	21	26	31	37	42	47	53	59	64	70	75	81	86	92	98
17	17	22	28	34	39	45	51	57	63	67	75	81	87	93	99	105
18	18	24	30	36	42	48	55	61	67	74	80	86	93	99	106	112
19	19	25	32	38	45	52	58	65	72	78	85	92	99	106	113	119
20	20	27	34	41	48	55	62	69	76	83	90	98	105	112	119	127

Critical Values of U at α = 0.05 with direction predicted or at α = 0.10 with direction not predicted.

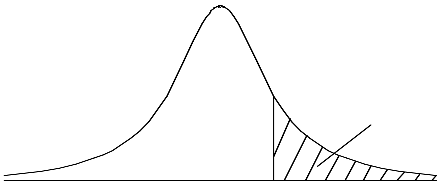
n_1 / n_2	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2	0	0	0	0	1	1	1	2	2	2	3	3	3	4	4	4
3	1	2	2	3	4	4	5	5	6	7	7	8	9	9	10	11
4	2	3	4	5	6	7	8	9	10	11	12	14	15	16	17	18
5	4	5	6	8	9	11	12	13	15	16	18	19	20	22	23	25
6	5	7	8	10	12	14	16	17	19	21	23	25	26	28	30	32
7	6	8	11	13	15	17	19	21	24	26	28	30	33	35	37	39
8	8	10	13	15	18	20	23	26	28	31	33	36	39	41	44	47
9	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54
10	11	14	17	20	24	27	31	34	37	41	44	48	51	55	58	62
11	12	16	19	23	27	31	34	38	42	46	50	54	57	61	65	69
12	13	17	21	26	30	34	38	42	47	51	55	60	64	68	72	77
13	15	19	24	28	33	37	42	47	51	56	61	65	70	75	80	84
14	16	21	26	31	36	41	46	51	56	61	66	71	77	82	87	92
15	18	23	28	33	39	44	50	55	61	66	72	77	83	88	94	100
16	19	25	30	36	42	48	54	60	65	71	77	83	89	95	101	107
17	20	26	33	39	45	51	57	64	70	77	83	89	96	102	109	115
18	22	28	35	41	48	55	61	68	75	82	88	95	102	109	116	123
19	23	30	37	44	51	58	65	72	80	87	94	101	109	116	123	130
20	25	32	39	47	54	62	69	77	84	92	100	107	115	123	130	138

Critical Values of U at α = 0.005 with direction predicted or at α = 0.01 with direction not predicted.

n_2 \ n_1	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	1	1	1	2	2	2	2	3	3
4	0	0	0	1	1	2	2	3	3	4	5	5	6	6	7	8
5	0	1	1	2	3	4	5	6	7	7	8	9	10	11	12	13
6	1	2	3	4	5	6	7	9	10	11	12	13	15	16	17	18
7	1	3	4	6	7	9	10	12	13	15	16	18	19	21	22	24
8	2	4	6	7	9	11	13	15	17	18	20	22	24	26	28	30
9	3	5	7	9	11	13	16	18	20	22	24	27	29	31	33	36
10	4	6	9	11	13	16	18	21	24	26	29	31	34	37	39	42
11	5	7	10	13	16	18	21	24	27	30	33	36	39	42	45	46
12	6	9	12	15	18	21	24	27	31	34	37	41	44	47	51	54
13	7	10	13	17	20	24	27	31	34	38	42	45	49	53	56	60
14	7	11	15	18	22	26	30	34	38	42	46	50	54	58	63	67
15	8	12	16	20	24	29	33	37	42	46	51	55	60	64	69	73
16	9	13	18	22	27	31	36	41	45	50	55	60	65	70	74	79
17	10	15	19	24	29	34	39	44	49	54	60	65	70	75	81	86
18	11	16	21	26	31	37	42	47	53	58	64	70	75	81	87	92
19	12	17	22	28	33	39	45	51	56	63	69	74	81	87	93	99
20	13	18	24	30	36	42	46	54	60	67	73	79	86	92	99	100

Table of statistics used for the calculation

Table of the Student's *t*-distribution



The table gives the values of $t_{\alpha;\nu}$ where

$\Pr(T_\nu > t_{\alpha;\nu}) = \alpha$, with ν degrees of freedom

α / ν	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.076	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646

α / ν	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Exercises
Chapter 2 Nonparametric Statistics

1. From a survey of the number of books borrowed from a certain library during the previous week, the following data were obtained.

	Monday	Tuesday	Wednesday	Thursday	Friday
Number of books borrowed (volumes)	135	108	120	114	146

Test the hypothesis whether the numbers of books borrowed on each day are different or not, using the 0.05 significance level.

2. In order to test whether family size depends on the educational level of the head of the family, a random sample of 200 married men who are old enough to no longer work was selected. It was found that each person with different educational levels has a family with the following numbers of children.

Educational level	Number of children of each person		
	0 – 1	2 - 3	More than 3
Primary education	14	36	30
Secondary education	18	42	20
Higher education	16	14	10

Using the 0.05 significance level, test whether family size depends on the educational level of the head of the family.

3. From an English subject examination that was graded using letter grades, it was found that males received grade A 30%, grade B 25%, and grade C 45%, while females received grade A 35%, grade B 20%, and grade C 45%. Test whether the ability in

learning English depends on gender, using the 0.05 significance level.

4. From a survey of 100 people who were smokers and non-smokers, it was found that among the 50 people who smoked, 8 people had cancer, and among the total of 86 people who did not have cancer, 44 were non-smokers. Test whether smoking affects cancer, using the 0.05 significance level.

5. From a study of the probabilities of first-year university students at a certain university four years ago, it was found that the probabilities of being under 17 years old, 17 to 18 years old, and over 18 years old were 0.05, 0.10, and 0.85, respectively. If a current random sample of 100 students from this university was selected and it was found that the numbers of students in each age group were 3, 8, and 89, respectively, test whether the probabilities of students being in each age group at present differ from those of four years ago, at the 0.05 significance level.

6. In surveying the popularity of a certain political party among the public, the researcher randomly selected eligible voters from Chiang Mai, Songkhla, Ubon Ratchathani, and Chonburi, 200 people from each province, and asked about their preference for this political party. The results were as follows:

	Province			
Favoring	Chiang Mai	Songkhla	Ubon Ratchathani	Chonburi
Favored	76	53	59	48
Unfavored	124	147	141	152

Test whether the popularity of the political party depends on the province in which the people reside, at the 0.05 significance level.

7. From a survey on whether the number of accidents that occurred is related to the age of car drivers, by surveying individuals aged 21 years and over during the past 3 months, the following data were obtained:

Number of accidents (times)	Age of car drivers				
	21 - 30	31 - 40	41 - 50	51 - 60	61 – 70
0	748	821	786	720	672
1	74	60	51	66	50
2	31	25	22	16	15
More than 2	9	10	6	5	7

Test whether the number of accidents that occur is related to the age of car drivers. Use $\alpha = 0.05$

Answers

Exercises for Chapter 2

1. $H_0: \pi_1 = 1/5, \pi_2 = 1/5, \pi_3 = 1/5, \pi_4 = 1/5, \pi_5 = 1/5$

   $H_1:$ There exists at least one π_i that is different from H_0

   Reject H_0 when $\chi^2_{cal} > 9.48773$ $\chi^2_{cal} = 7.8267$ Accept H_0

2. $H_0:$ Family size does not depend on the educational level of the head of the family.

   $H_1:$ Family size depends on the educational level of the head of the family. Reject H_0 when $\chi^2_{cal} > 9.48772$ $\chi^2_{cal} = 10.0544$

   Reject H_0

3. $H_0:$ Ability in learning English does not depend on gender.

   $H_1:$ Ability in learning English depends on gender.

   Critical value = 5.99147 $\chi^2_{cal} = 0.9402$ Accept H_0

4. $H_0:$ Smoking has no effect in causing cancer.

   $H_1:$ Smoking has an effect in causing cancer.

   Reject H_0 when $\chi^2_{cal} > 3.84146$ $\chi^2_{cal} = 0.3324$ Accept H_0

5. $H_0 : \pi_1 = 0.05, \pi_2 = 0.10, \pi_3 = 0.85$

H_1: There exists at least one π_i that is different from the theoretical value in H_0

Reject H_0 when $\chi^2_{cal} > 5.99147$ $\chi^2_{cal} = 1.3882$ Accept H_0

6. H_0: The popularity of the political party among the public does not depend on the province in which the people reside.

H_1: The popularity of the political party among the public depends on the province in which the people reside.

Reject H_0 when $\chi^2_{cal} > 7.81473$ $\chi^2_{cal} = 10.7224$ Reject H_0

7. H_0: The number of accidents that occur is not related to the age of the car drivers.

H_1: The number of accidents that occur is related to the age of the car drivers.

Reject H_0 when $\chi^2_{cal} > 21.0261$ $\chi^2_{cal} = 14.349$ Accept H_0